Accepted version, final version available at http://arc.aiaa.org/doi/abs/10.2514/1.G009427

Robust Design of Low-Thrust Gravity-Assist Trajectories via Reinforcement Learning

Jincheng Hu*, Hongwei Yang†, Shuang Li‡
Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China
and
Hexi Baoyin§
Tsinghua University, Beijing 100084, China

Abstract

This paper presents a reinforcement-learning (RL) based multi-phase robust trajectory design method for low-thrust exploration with gravity assist (GA) under uncertainties. To alleviate the complexity due to the interior-point constraints caused by intermediate GA, a trajectory segmentation strategy is used. The low-thrust trajectory legs before and after GA are segmented into interplanetary transfer phases (ITPs) and approaching phases (APs), respectively, and Markov decision processes with stochastic dynamics are modeled in each phase. Moreover, for the issue of failing to reach the target state of the AP due to the low terminal accuracy of the preceding ITP, a trade-off factor is defined to modify the nominal initial state of the AP to better guide the exploration of robust policies. In particular, a reachability constraint of the target state of the AP is modeled analytically and incorporated into the reward of the ITP, which can significantly improve the reachability from the end states of RL-based trajectories in the preceding ITP to the target of the AP. Besides, low-thrust robust guidance laws in the AP are trained to deal with the uncertainties over the AP. The promising results in an Earth-Earth-Jupiter mission show that the proposed method can not only effectively deal with various uncertainties, but also achieve the desired accuracy for intermediate GA and terminal rendezvous.

I. Introduction

* PhD Candidate, College of Astronautics; hujincheng@nuaa.edu.cn.
† Professor, College of Astronautics; hongwei.yang@nuaa.edu.cn. Senior Member AIAA (Corresponding author).
‡ Professor, College of Astronautics; lishuang@nuaa.edu.cn. Senior Member AIAA
§ Professor, School of Aerospace Engineering; baoyin@tsinghua.edu.cn. Senior Member AIAA.

The electric propulsion (EP) technology is becoming increasingly used in deep space exploration missions [1]. EP systems present the attractive properties of high specific impulse and lower propellant consumption, but also pose challenges to low-thrust trajectory optimization. Meanwhile, low-thrust trajectories are subject to various uncertainties over the mission, e.g., state uncertainties, observation uncertainties, control uncertainties, and extreme situations of missed thrust, all of which threaten the successful execution of the mission [2] - [4]. Besides, gravity assist (GA) is an important technique for shortening flight time and saving fuel in low-thrust deep space missions [5] - [9], especially for reaching the outer planets. The well-known Jovian system tour missions [10] [11] also rely on a series of GA maneuvers. The GA maneuver imposes complex interior-point constraints on low-thrust trajectories, making the optimal control problems more challenging. Due to the weak orbital correction capabilities of EP spacecraft and communication delays in deep space, it is highly desirable for the spacecraft to be capable of addressing unexpected uncertainties autonomously during a mission [12] [13].

The conventional methods for low-thrust trajectory optimization, such as indirect and direct methods [14] - [19], are mostly formulated in deterministic scenarios and not applicable to robust trajectory design problems. Typically, robust trajectories are obtained by adding margins in the deterministic system, but it takes time-consuming iterations and often leads to over-conservative solutions [20]. Recently, many numerical methods have been developed for robust low-thrust trajectory design. Ozaki et al. [21] [22] applied stochastic differential dynamic programming and unscented transform to robust low-thrust trajectory design considering Gaussian-distributed state noise. In [23], a direct transcription method integrating multiple shooting with polynomial algebra was proposed, where the stochastic and epistemic uncertainties were incorporated. Oguri et al. [24] developed stochastic prime vector theory in the field of constrained optimal orbit transfer under state uncertainties, by which the behavior of optimal trajectories under uncertainties and path constraints was clarified. Meanwhile, convexification technique is applied to low-thrust trajectory optimization under Wiener process [25].

Among these methods, prior tailored transformations are necessitated to recast originally intractable problems into specific paradigms. For scenarios considering control uncertainties, they are not directly applicable as such disturbances cannot be modeled as additive Gaussian noise [2]. It is also worth noting that GA maneuver is rarely considered in these studies.

Besides, with the development in machine learning (ML), RL is emerging as an alternative to traditional optimal control methods in astrodynamics applications. In RL, the problem is formulated as a Markov decision process (MDP), characterizing the interaction between the agent and the environment. Through iterative interactions, an agent explores the optimal policy by maximizing the expected cumulative sum of rewards [26]. Currently, RL has been widely applied in astrodynamics problems, e.g., robust low-thrust trajectory design [2] [4], station-keeping maneuvers [27], and autonomous guidance in scenarios including asteroid approaching [28], lunar transfer [29], asteroid impact [30], and planetary and asteroid landing [31] [32]. Zavoli et al. [2] firstly systematically investigated the robust low-thrust trajectory design problems for interplanetary rendezvous under various uncertainties via RL, and promising results were obtained. Nevertheless, to the best of our knowledge, there has been no application of RL to robust low-thrust GA trajectory design. It is mainly due to the low terminal accuracy of RL-based interplanetary transfer trajectories [2] [4], coupled with the inability of EP systems to compensate for large state deviations within the limited remaining time [33]. As such, the additional interior-point constraints caused by GA, due to the requirement to match the planet's position at encounter, cannot be satisfied.

In a preliminary study, the impulsive GA model is a common practice, where the GA maneuver is modeled as an instantaneous change in heliocentric velocity [8] [9]. To achieve the desired GA maneuver, the spacecraft must satisfy specific positional constraints within the scheduled GA window. These interior-point constraints pose significant challenges to low-thrust GA trajectory design. Carnelli et al. [6] developed an innovative method for low-thrust GA trajectory optimization based on evolutionary neurocontrollers, which shared a similar underlying principle to RL. Nevertheless, uncertainties were not considered, and this method also suffered from low terminal accuracy in trajectory design. In [8], a low-thrust GA fuel-optimal trajectory optimization problem with planetary and solar radiation pressure perturbations was investigated, but it was still a deterministic problem under an accurately known dynamical model. Ozaki et al. [21] incorporated unscented transform into stochastic differential dynamic programming, and applied it to the robust low-thrust trajectory design of the V-Infinity leveraging problem under state uncertainties. The resulting terminal miss distance was on the order of 10^5 km, rendering it infeasible to generate the intended GA maneuver. Venigalla et al. [33] proposed a novel virtual swarm technique to enhance the robustness of low-thrust trajectory to missed thrust event, while only a single missed thrust event was accounted for, and other uncertainties were not considered.

In summary, the incorporation of GA in robust low-thrust trajectory design will further complicate the tailored transformation process required by existing numerical methods [22] - [25], thereby making robust trajectory design more challenging. RL-based methods [2] [4] face the issue of insufficient terminal accuracy in interplanetary transfer to meet the interior-point constraints induced by GA. In [28], a robust zero-effort-miss/zero-effort-velocity (R-ZEM/ZEV) method was proposed for low-thrust approaching under uncertainties, achieving high terminal accuracy. It is worth noting that R-ZEM/ZEV [28] is only suitable for short-arc approaching problems, as its fuel-optimal performance will degrade significantly over long-arc transfers [34]. Cheng et al. [35] designed a multiscale cascaded network for the highly precise control of terminal orbit insertion, while only the time-optimal coplanar transfer problem with initial state uncertainties was considered. The network scale would be more complex for non-coplanar transfer problems, and incorporating process uncertainties would pose challenges to the pre-data generation required by the method in [35].

In this paper, an RL-based multi-phase robust GA trajectory design method is proposed for low-thrust exploration missions with intermediate GA under uncertainties. Specifically, the innovation work is mainly in the following three folds. Firstly, a trajectory segmentation strategy is used to alleviate the complexity due to the interior-point constraints caused by GA, by which the low-thrust trajectory legs before and after GA are segmented into ITPs and APs, respectively. In each phase, the MDPs are modeled separately for the considered uncertainties. Different from the multiscale network strategy in [35], different mapping schemes are formulated for the ITPs and APs in this study, i.e., the mapping of state-velocity increment is for the former and the state-guidance parameters in the R-ZEM/ZEV method [28] for the latter. This enables the robust low-thrust trajectory design problem with GA to be addressed using simpler network scales. Secondly, to mitigate the issue of the EP spacecraft failing to reach the target state of the AP due to significant end-state deviations from the preceding ITP, a trade-off factor is defined to modify the nominal initial state of the AP, thereby reducing the required velocity increment over the AP. This modified nominal initial state of the AP can better guide the exploration of robust policies in RL training, by steering the disturbed EP spacecraft toward states requiring lower velocity increments to reach the target state of the AP. Thirdly, an analytical reachability constraint of the target of the AP is modeled and incorporated into the reward of the ITP, remarkably enhancing the reachability from the end states of RL-based ITP trajectories to the target of the AP. On this basis, the low-thrust robust guidance laws can be applied to address uncertainties over the AP and achieve high approaching accuracy.

The rest of the paper is organized as follows. In Section II, the investigated robust low-thrust GA trajectory design problem and the GA model are stated. Subsequently, Section III details the proposed RL-based multi-phase robust trajectory design method. Thereafter, the performance of the proposed method is validated in an Earth-Earth-Jupiter mission. Finally, Section V concludes the paper.

II. Problem Background

A. Dynamics of Low-Thrust Trajectories

In this study, a preliminary problem related to deep space rendezvous missions with intermediate GA is investigated, where the spacecraft is equipped with an EP system. In the mission, it is assumed that the spacecraft is mainly subject to the central force of the Sun's gravity and the EP engine's thrust. The well-known Sims-Flanagan (S-F) transcription [36] is employed to model the dynamics of the EP spacecraft. Using the S-F, the continuous low-thrust trajectory of each phase is discretized into N_i ballistic arcs connected by impulses $\Delta V_{i,k}$ at the k-th discrete points $t_{i,k} = ktof_i / N_i$, $k \in [0, N_i]$. The term tof_i denotes the duration of the i-th discretized trajectory phase, where the subscript i indicates the number of the trajectory phase.

For the (deterministic) unperturbed scenario, the state propagation equations of the EP spacecraft are

$$\begin{bmatrix} \boldsymbol{r}_{i,k+1} \\ \boldsymbol{v}_{i,k+1} \end{bmatrix} = \begin{bmatrix} \hat{f}_{i,k}\boldsymbol{r}_{i,k} + \hat{g}_{i,k}\left(\boldsymbol{v}_{i,k} + \Delta\boldsymbol{V}_{i,k}\right) \\ \dot{\hat{f}}_{i,k}\boldsymbol{r}_{i,k} + \dot{\hat{g}}_{i,k}\left(\boldsymbol{v}_{i,k} + \Delta\boldsymbol{V}_{i,k}\right) \end{bmatrix} = F(\boldsymbol{r}_{i,k}, \boldsymbol{v}_{i,k}, \Delta\boldsymbol{V}_{i,k}) \tag{1}$$

$$m_{i,k+1} = m_{i,k} \exp\left(-\left\|\Delta\boldsymbol{V}_{i,k}\right\| / u_{eff}\right) \tag{2}$$

where $\boldsymbol{r}_{i,k}$, $\boldsymbol{v}_{i,k}$ are the position and velocity vector of the spacecraft in heliocentric ecliptic reference framework, and $m_{i,k}$ is its mass. The function $\boldsymbol{F}(\boldsymbol{r}_{i,k}, \boldsymbol{v}_{i,k}, \Delta \boldsymbol{V}_{i,k})$ denote the well-known Lagrangian coefficient formula, and $\hat{f}_{i,k}$ and $\hat{g}_{i,k}$ are the Lagrange's coefficients, defined as in Ref. [37]. An astrodynamics library *pykep* [38] is used for the state propagation in Eq. (1). The Tsiolkovsky equation (2) is adopted to update the mass, and $u_{eff} = g_0 I_{sp}$ is the effective exhaust velocity, where I_{sp} is the specific impulse of the EP engine and $g_0 = 9.80665$ m/s^2. In particular, the velocity increment $\Delta\boldsymbol{V}_{i,k}$ is an approximate equivalent effect of the low-thrust acceleration over the discrete arc, which is limited by the cumulative effect $\Delta V_{i,k}^{\max}$ of the maximum thrust T_{max} along the arc:

$$\Delta V_{i,k}^{\max} = \frac{T_{\max}}{m_{i,k}} \frac{tof_i}{N_i} \qquad \Delta\boldsymbol{V}_{i,k} \in \left[-\Delta V_{i,k}^{\max}, \Delta V_{i,k}^{\max}\right]^3 \subset \mathbb{R}^3 \tag{3}$$

Meanwhile, there are various non-negligible uncertainties during deep space missions. Due to the weak trajectory-correction capability of the EP spacecraft, it is necessary to take into account the uncertainties when designing low-thrust trajectories. The considered uncertainties in this work include state uncertainties due to the idealized kinetic model, observation uncertainties from measurement noises, control uncertainties caused by mechanical errors, and extreme cases of missed thrust events.

As in [2] [4], the state uncertainties are modeled as zero-mean additive Gaussian noise on the position and velocity of the spacecraft at time t_k, $k \in (0, N_i]$, denoted as $\delta \boldsymbol{r}_{i,k}^{st}$ and $\delta \boldsymbol{v}_{i,k}^{st}$, respectively.

$$\begin{bmatrix} \delta \boldsymbol{r}_{i,k}^{st} \\ \delta \boldsymbol{v}_{i,k}^{st} \end{bmatrix} \sim \mathcal{N}\left(\boldsymbol{0}_6, \boldsymbol{R}_{i,s}\right), \quad \boldsymbol{R}_{i,s} = \begin{bmatrix} \sigma_{i,rs}^2 \boldsymbol{I}_3 & \boldsymbol{0}_3 \\ \boldsymbol{0}_3 & \sigma_{i,vs}^2 \boldsymbol{I}_3 \end{bmatrix} \tag{4}$$

where $\boldsymbol{R}_{i,s}$ is the covariance matrix, with $\sigma_{i,rs}$ and $\sigma_{i,vs}$, the standard deviations on position and velocity, and $\boldsymbol{I}_n$ (respectively, $\boldsymbol{0}_n$) is an n-dimensional identity (null) matrix. The stochastic dynamic model of the spacecraft is

$$\begin{bmatrix} \boldsymbol{r}_{i,k+1} \\ \boldsymbol{v}_{i,k+1} \\ m_{i,k+1} \end{bmatrix} = \begin{bmatrix} F(\boldsymbol{r}_{i,k}, \boldsymbol{v}_{i,k}, \Delta \boldsymbol{V}_{i,k}) \\ m_{i,k} \exp\left(-\dfrac{\left\|\Delta \boldsymbol{V}_{i,k}\right\|}{u_{eq}}\right) \end{bmatrix} + \begin{bmatrix} \delta \boldsymbol{r}_{i,k}^{st} \\ \delta \boldsymbol{v}_{i,k}^{st} \\ 0 \end{bmatrix} \tag{5}$$

The observation uncertainties are also considered to be zero-mean additive Gaussian noise on the knowledge of spacecraft's position and velocity, being $\delta \boldsymbol{r}_{i,k}^{obs}$ and $\delta \boldsymbol{v}_{i,k}^{obs}$, respectively.

$$\begin{bmatrix} \delta \boldsymbol{r}_{i,k}^{obs} \\ \delta \boldsymbol{v}_{i,k}^{obs} \end{bmatrix} \sim \mathcal{N}\left(\boldsymbol{0}_6, \boldsymbol{R}_{i,o}\right) \tag{6}$$

where $\boldsymbol{R}_{i,o} = \mathrm{diag}(\sigma_{i,ro}^2 \boldsymbol{I}_3, \sigma_{i,vo}^2 \boldsymbol{I}_3)$ is the corresponding covariance matrix, $\sigma_{i,ro}$ and $\sigma_{i,vo}$ are the standard deviations on the observed position and velocity of the EP spacecraft.

For control uncertainties, the Gaussian control magnitude error $\delta u_{i,k} \sim \mathcal{N}(0, \sigma_{i,u}^2)$ and Euler angle error $(\delta\phi_{i,k}, \delta\vartheta_{i,k}, \delta\psi_{i,k}) \sim \mathcal{N}(\boldsymbol{0}_3, \sigma_{i,Eu}^2 \boldsymbol{I}_3)$ are considered. The resulting actual control $\boldsymbol{u}_{i,k}$ is [4]

$$\boldsymbol{u}_{i,k} = (1 + \delta u_{i,k}) \begin{bmatrix} 1 & -\delta\psi_{i,k} & \delta\vartheta_{i,k} \\ \delta\psi_{i,k} & 1 & -\delta\phi_{i,k} \\ -\delta\vartheta_{i,k} & \delta\phi_{i,k} & 1 \end{bmatrix} \Delta \boldsymbol{V}_{i,k} \tag{7}$$

As for missed thrust events, they represent extreme cases where the actual control $\boldsymbol{u}_{i,k} = \boldsymbol{0}_3$ at k-th step due to the breakdown or safe mode of the EP system [39]. In this study, two types of missed thrust events are considered. One is that the breakdown is assumed to happen only once at a random step $t_{i,k}$, $k \in [0, N_i)$, labeled as *mte*1. The other (*mte*3) is that miss-thrust condition may recovery with the probability $P_{i,mte}$ at next step $t_{i,k+1}$, otherwise it will persist for an addition step, and this process may last for up to $N_{i,\,mte}$ consecutive steps [2] [4].

B. Gravity Assist Formulation

For the intermediate GA maneuvers, the GA impulse model is adopted, as in [8] [14]. Specifically, the spacecraft's positions immediately before and after the GA are consistent with the planet position, while the hyperbolic excess velocity is rotated by an angle δ with its magnitude remains constant [14].

$$\begin{cases} t_G^- = t_G^+ = t_G \\ \boldsymbol{r}(t_G^-) = \boldsymbol{r}(t_G^+) = \boldsymbol{r}_{\mathrm{pl}}(t_G) \\ \boldsymbol{v}(t_G^-) = \boldsymbol{v}_{\mathrm{pl}}(t_G) + \boldsymbol{v}_\infty^- \\ \boldsymbol{v}(t_G^+) = \boldsymbol{v}_{\mathrm{pl}}(t_G) + \boldsymbol{v}_\infty^+ \\ \left\| \boldsymbol{v}_\infty^- \right\| = \left\| \boldsymbol{v}_\infty^+ \right\| = v_\infty \end{cases} \tag{8}$$

where t_G is the time of GA, the superscripts '-' and '+' indicate the time instants immediately before and after GA, $\boldsymbol{r}_{\mathrm{pl}}$ and $\boldsymbol{v}_{\mathrm{pl}}$ denote the position and velocity of the planet, respectively. Meanwhile, $\boldsymbol{v}_\infty^-$ and $\boldsymbol{v}_\infty^+$ are the incoming and outgoing hyperbolic excess velocities, and the GA maneuver is generated from the rotation of $\boldsymbol{v}_\infty^-$ towards $\boldsymbol{v}_\infty^+$ under the planet's gravitational influence. For a GA maneuver, the turn angle δ between $\boldsymbol{v}_\infty^-$ and $\boldsymbol{v}_\infty^+$ depends on the spacecraft's periapsis radius r_p with respect to the planet and on the magnitude of the hyperbolic excess velocity v_∞.

$$\delta = 2\arcsin\left(\frac{1}{1+\left(r_p/\mu_{\mathrm{pl}}\right)v_\infty^2}\right) \tag{9}$$

where μ_{Pl} denotes the gravitational constant of the GA planet. From Eq. (9), it can be noticed that the maximum turn angle is limited by the minimal admissible periapsis radius $r_{p,\,\min}$, which is usually prescribed in advance.

Based on Eqs. (8) and (9), the low-thrust trajectories before and after GA can be spliced. In this work, the optimal GA maneuver is calculated using the indirect method [8] [14] before the RL training of robust policies.

C. Robust Gravity-Assist Trajectory Design Problem

The robust low-thrust trajectory design with GA is to find a series of $\Delta \boldsymbol{V}_{i,\,k}$, that satisfies the interior-point constraints in Eq. (8) and the terminal rendezvous constraint, while maximizing the final mass:

$$\boldsymbol{r}(t_f) = \boldsymbol{r}_{tgt}, \quad \boldsymbol{v}(t_f) = \boldsymbol{v}_{tgt} \tag{10}$$

where t_f is the final time of the whole mission, and ($\boldsymbol{r}_{tgt}$, $\boldsymbol{v}_{tgt}$) denotes the states of the rendezvous target at t_f. The uncertainties mentioned in Sect. II. A are considered separately in this study, and the EP spacecraft is subject to the corresponding stochastic equations of motion described therein.

Different from the widely studied robust low-thrust trajectory design without GA [2], [4], [22] - [25], the intermediate GA induces stringent interior-point constraints into trajectory design, notably, the matching of the position with the flyby body at time t_G. The interior-point constraints render existing methods inapplicable to the investigated problem, especially for the cases with multiple GA maneuvers. Meanwhile, the nonlinear propagation of uncertainties makes these constraints more intractable in robust trajectory design [22]. Furthermore, considering the communication delay in deep space, it is of high interest that the EP spacecraft is capable of addressing uncertainties autonomously during a mission. For this, RL is applied to the investigated problem, and the robust policies are trained for the considered uncertain scenarios, as detailed in Sect. III.

III. Reinforcement-Learning Based Multi-Phase Robust Trajectory Design Method

A. Low-Thrust Trajectory Segmentation Strategy

As mentioned, the interior-point constraints induced by GA and the nonlinear propagation of the uncertainties pose challenges to the design of robust low-thrust GA trajectories. For the state-velocity increment mappings established by the RL-based methods in Refs. [2] [4], the terminal accuracy of the solutions is insufficient to meet the desired accuracy for GA maneuvers and terminal rendezvous. For this, an RL-based multi-phase robust low-thrust GA trajectory design method is developed. Overall, the low-thrust trajectory legs before and after a GA event are divided into ITPs and APs, respectively, as shown in Fig. 1. In this mission, the EP spacecraft departs from Earth, returns after a duration exceeding Earth's orbital period, performs a GA maneuver, and ultimately reaches Jupiter. The whole mission trajectory is divided into 2(n+1) phases, where n is the number of GA maneuvers. The trajectory phases are numbered sequentially, with the ITPs always corresponding to the odd-numbered phases and the APs to the even-numbered. As well, the MDPs for the different trajectory phases are formulated separately, with boundary conditions determined via the indirect methods [8] [14]. It is worth noting that the slight differences between the solutions obtained by the indirect methods and those from the direct method with the S-F model have a negligible impact on the proposed method.

The trajectory segmentation strategy effectively alleviates the complexity of the problem due to the interior-point constraints caused by GA. This strategy is also suitable for multiple GA missions, by which the complexity of the problem grows linearly with the number of GA maneuvers. In particular, the R-ZEM/ZEV method [28] proposed by the authors is applied to generate robust guidance laws for the APs, to achieve a high terminal accuracy for the interior-point constraints in Eq. (8) and the rendezvous constraints in Eq. (10).

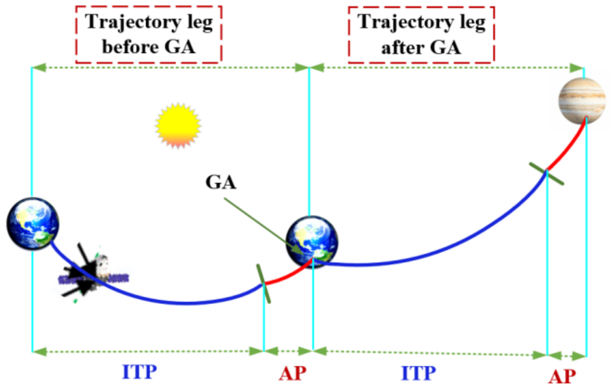


Fig. 1 The segmentation schematic of low-thrust gravity-assist trajectory

B. Markov Decision Processes Formulation

To comply with the standard formulation of RL, the robust trajectory design problem needs to be recast into a generic MDP, which characterizes the agent-environment interaction in RL. In general, the MDP consists of a state space $\boldsymbol{S}$, an action space $\boldsymbol{U}$, a dynamics-dependent state transition probability distribution $\boldsymbol{P}(\boldsymbol{s}_{i,k+1}|\boldsymbol{s}_{i,k}, \boldsymbol{a}_{i,k})$, and a reward function $r_{i,k}$, where $\boldsymbol{s}_{i,k} \in \boldsymbol{S}$ and $\boldsymbol{a}_{i,k} \in \boldsymbol{U}$ [4] [28]. The subscript i indicates the MDPs for different trajectory phases, as the MDPs for ITPs and APs are formulated separately due to their distinct exploration strategies for orbital maneuvers. In RL training, an agent takes an action $\boldsymbol{a}_{i,\,k}$ according to the observation $\boldsymbol{o}_{i,\,k}$ and the current policy $\pi(\boldsymbol{a}_{i,\,k}|\,\boldsymbol{o}_{i,\,k})$, updates its state based on $\boldsymbol{P}(\boldsymbol{s}_{i,\,k+1}|\boldsymbol{s}_{i,\,k},\,\boldsymbol{a}_{i,\,k})$, and receives a scalar reward. By interacting repeatedly with the environment, an agent collects the batch information and updates the current policy by maximizing the expected cumulative sum of rewards over an episode.

1. Interplanetary Transfer Phases

For the ITPs, the state-velocity increment mappings are explored and established for robust low-thrust trajectory design, as in Refs. [1] [4]. In this study, both the unperturbed deterministic scenario and uncertain scenarios are considered. For the unperturbed scenario (labeled as *unp*), the agent state $\boldsymbol{s}_{i,\,k}$ is composed of the spacecraft's position $\boldsymbol{r}_{i,\,k}$, velocity $\boldsymbol{v}_{i,\,k}$, and the mass $m_{i,\,k}$:

$$\boldsymbol{s}_{i,\,k}=\left[\boldsymbol{r}_{i,\,k}^T,\ \boldsymbol{v}_{i,\,k}^T,\ m_{i,\,k}\right]^T\in\mathbb{R}^7 \tag{11}$$

The agent action $\boldsymbol{a}_{i,\,k}$ at the k-th step is a three-dimensional vector: $\boldsymbol{a}_{i,\,k}=\Delta\boldsymbol{V}_{i,\,k}$, *s.t.* ($\left\|\boldsymbol{a}_{i,k}\right\|\le\Delta V_{i,\,k}^{\max}$). The agent states are updating based on the Lagrangian coefficient formula [37] and Tsiolkovsky's equation, as mentioned in Sect. II. A.

For uncertain scenarios, a fuel-optimal trajectory solved by the indirect method [14] in the unperturbed scenario *unp* is adopted as a reference trajectory, by which the isochronous reference kinetic states at each step of the spacecraft can be provided to the agent. The agent state $\boldsymbol{s}_{i,\,k}$ in these scenarios incorporates not only the actual state of the spacecraft in Eq. (11), but also the isochronous reference states, i.e., the position $\boldsymbol{r}_{i,\,ref(k)}$ and velocity $\boldsymbol{v}_{i,\,ref(k)}$, enabling the agent to explore the robust policies more efficiently.

$$\boldsymbol{s}_{i,k}=\left[\boldsymbol{r}_{i,k}^T,\boldsymbol{v}_{i,k}^T,m_{i,k},\boldsymbol{r}_{i,ref(k)}^T,\boldsymbol{v}_{i,ref(k)}^T\right]^T\in\mathbb{R}^{13} \tag{12}$$

In the scenarios where state uncertainties are considered (*st*), the actual state of the EP spacecraft is perturbed by the Gaussian noises $\delta\boldsymbol{r}_{i,k}^{st}$ and $\delta\boldsymbol{v}_{i,k}^{st}$ in Eq. (4), and they are updated via Eq. (5). The isochronous reference state ($\boldsymbol{r}_{i,\,ref(k)}$, $\boldsymbol{v}_{i,\,ref(k)}$) at k-th step is directly taken from the reference trajectory.

For the scenarios where Gaussian observation noises ($\delta \boldsymbol{r}_{i,k}^{obs}$ and $\delta \boldsymbol{v}_{i,k}^{obs}$) are included in the knowledge of spacecraft's position and velocity (*obs*), the corresponding observation $\boldsymbol{o}_{i,k}$ available to the agent is [4]

$$\boldsymbol{o}_{i,k} = \begin{bmatrix} \boldsymbol{r}_{i,k} \\ \boldsymbol{v}_{i,k} \\ m_{i,k} \\ \boldsymbol{r}_{i,ref(k)} \\ \boldsymbol{v}_{i,ref(k)} \end{bmatrix} + \begin{bmatrix} \delta \boldsymbol{r}_{i,k}^{obs} \\ \delta \boldsymbol{v}_{i,k}^{obs} \\ \boldsymbol{0}_1 \\ \boldsymbol{0}_3 \\ \boldsymbol{0}_3 \end{bmatrix} \tag{13}$$

As for the scenarios considering control uncertainties (*ctr*) and the extreme scenarios of *mte*1 and *mte*3, the state of the EP spacecraft ($\boldsymbol{r}_{i,k}$, $\boldsymbol{v}_{i,k}$, $m_{i,k}$) is updated by substituting the actual control $\boldsymbol{u}_{i,k}$ in Eqs. (1) and (2).

2. Approaching Phases

Next, the MDPs for the APs are modeled. Similarly, both the unperturbed scenario *unp* and uncertain scenarios (i.e., *st*, *obs*, *ctr*, *mte*1, *mte*3) are investigated. Different from the ITPs, the mappings between the agent states and guidance parameters are explored for the robust guidance over the APs. The R-ZEM/ZEV [28] method is employed to generate low-thrust robust guidance laws over the APs to deal with uncertainties, owing to its simplicity and fuel near-optimality in the short-arc approaching problem [28] [34]. Due to the short-arc properties, there is no need for a reference trajectory in the APs. Referring to Ref. [28], the agent state $\boldsymbol{s}_{i,k}$ is

$$\boldsymbol{s}_{i,k} = \left[(\boldsymbol{r}_{i,k} - \boldsymbol{r}_{i,k}^{tgt})^T,\ (\boldsymbol{v}_{i,k} - \boldsymbol{v}_{i,k}^{tgt})^T,\ m_{i,k},\ t_{i,k} \right]^T \in \mathbb{R}^8 \tag{14}$$

where $\boldsymbol{r}_{i,k}^{tgt}$ and $\boldsymbol{v}_{i,k}^{tgt}$ are the isochronous position and velocity of the target state in an AP. It is based on the assumption that there is a zero-control virtual spacecraft, whose terminal state at the end of the AP is the same as the target state of the current phase (i.e., target position $\boldsymbol{r}_i^{tgt}$ and velocity $\boldsymbol{v}_i^{tgt}$). Thus, the APs can be regarded as a rendezvous process between the EP spacecraft and the virtual spacecraft. The setting in Eq. (14) increases the differences of the agent states between the discretized steps $t_{i,k}$ over the short-arc approaching, and facilitates the establishment of mappings for robust guidance laws. In MDPs, the target state ($\boldsymbol{r}_i^{tgt}$, $\boldsymbol{v}_i^{tgt}$) of the current phase is determined in advance via the indirect method [8] [14], which is also for that of the ITPs. For the agent state update, the state ($\boldsymbol{r}_{i,k}^{tgt}$, $\boldsymbol{v}_{i,k}^{tgt}$) can be obtained by back-propagating ($\boldsymbol{r}_{i}^{tgt}$, $\boldsymbol{v}_{i}^{tgt}$) to the current step $t_{i,k}$ under simple two-body dynamics. The position $\boldsymbol{r}_{i,k}$ and velocity $\boldsymbol{v}_{i,k}$ of the spacecraft are updated via the formula $F(\boldsymbol{r}_{i,k}, \boldsymbol{v}_{i,k}, \Delta \boldsymbol{V}_{i,k})$ in Eq. (1) in scenario *unp*, the mass $m_{i,k}$ is updated by Eq. (2), and the timestamp is updated by $t_{i,k+1} = t_{i,k} + tof_i / N_i$. In particular, the velocity increment $\Delta \boldsymbol{V}_{i,k}$ is calculated using the classical ZEM/ZEV feedback guidance laws [34]:

$$\boldsymbol{a}_{c,i,k} = \frac{K_R}{t_{go}^2}\mathbf{ZEM} + \frac{K_V}{t_{go}}\mathbf{ZEV}, \quad t_{go} = tof_i - t_{i,k} \tag{15}$$

$$\Delta \mathbf{V}_{i,k} = sat_{\Delta V_{i,k}^{\max}}\left(\boldsymbol{a}_{c,i,k}\frac{tof_i}{N_i}\right), \quad sat_U = \begin{cases} \boldsymbol{q} & \text{if } \|\boldsymbol{q}\| \le U \\ U\boldsymbol{q}/\|\boldsymbol{q}\| & \text{if } \|\boldsymbol{q}\| > U \end{cases} \tag{16}$$

where K_R and K_V are the guidance parameters, $\boldsymbol{a}_{c,i,k}$ is the control acceleration in the ZEM/ZEV guidance laws. The terms **ZEM** and **ZEV** are the differences between the desired terminal state ($\boldsymbol{r}_{i}^{tgt}$, $\boldsymbol{v}_{i}^{tgt}$) of this phase and the projected final state of the spacecraft with no control effort (i.e., $\boldsymbol{r}_i^{nc}$, $\boldsymbol{v}_i^{nc}$, where *nc* is the abbreviation of no control) from current time $t_{i,k}$ onward: $\mathbf{ZEM} = \boldsymbol{r}_i^{tgt} - \boldsymbol{r}_i^{nc}$, $\mathbf{ZEV} = \boldsymbol{v}_i^{tgt} - \boldsymbol{v}_i^{nc}$. The normalized saturation function sat_U of a vector $\boldsymbol{q}$ indicates the low-thrust control magnitude constraint, as in Eq. (3). It follows that the control command is calculated based on the deviation between the actual terminal state and the target in ZEM/ZEV method, reflecting its feedback characteristics. This guidance law has mainly been applied in space scenarios such as intercept or rendezvous missions, terminal guidance, and planetary or lunar landing, where the guidance parameters are derived analytically: K_R = 6, K_V = -2 [34]. It is worth noting that although the feedback characteristics facilitate robustness against uncertainties, this is contingent upon the required velocity increment not exceeding the maximum magnitude $\Delta V_{i,k}^{\max}$. For low-thrust missions under uncertainties, the saturation function in Eq. (16) is easily triggered due to the low allowable control magnitude, resulting in a truncation of the desired velocity increment and poor terminal accuracy. Thereby, it is necessary to further explore the control gains (K_R, K_V), so that the guidance laws can guide the EP spacecraft to reach the target state even in the presence of uncertainties. However, when the gains are directly explored in RL, as in [32], the results are unsatisfactory [28] [32]. It is due to the fact that the saturation constraint is easily activated in this learning manner, and the resulting poor terminal accuracy will destabilize the learning process [28]. This makes it difficult for the agent to effectively update its policies during RL training. To this end, a method derived from Eq. (15), i.e., R-ZEM/ZEV has been proposed in Ref. [28], which will be briefly introduced next.

Considering the short-arc characteristic of the AP, it is assumed that gravitational acceleration remains constant throughout an AP. Then, a ZEM/ZEV feedback system can be formulated [28] [32]:

$$\dot{\mathbf{ZEM}} = -\boldsymbol{a}_{c,i,k} t_{go}, \ \dot{\mathbf{ZEV}} = -\boldsymbol{a}_{c,i,k} \tag{17}$$

where the dot at the top means the derivatives with respect to time. It is worth noting that this assumption is only applied in stability analysis of the feedback system, where the actual projected final state ($\boldsymbol{r}_i^{nc}, \boldsymbol{v}_i^{nc}$) is obtained from the state propagation as in Eqs. (1) and (5). For this feedback system, the eigenvalues can be derived via the linear time-varying system theorems [40]:

$$\lambda_{1,2} = \frac{-K \pm \sqrt{\Delta}}{2}, \quad K = K_R + K_V + 1, \quad \Delta = K^2 - 4K_R \tag{18}$$

According to Lyapunov stability theory, a system is stable only when the real parts of its eigenvalues are strictly less than zero. The stability behavior of this system can be divided into two cases depending on the value of Δ: oscillation-free stabilization if $\Delta \geq 0$ and oscillatory for $\Delta < 0$. Since the original control gains ($K_R = 6$, $K_V = -2$) are indeed compatible with $\Delta \geq 0$ and the oscillation-free stabilization is preferred, only the former case is considered in R-ZEM/ZEV method. From Eq. (18), an eigenvalue-related term Λ is defined:

$$\Lambda = K - \sqrt{\Delta} = K - \sqrt{K^2 - 4K_R} \tag{19}$$

It can be noted that the stability of the ZEM/ZEV system can be identified by the term Λ alone, and directly exploring Λ can significantly improve the stability of the RL training process [28]. Particularly, combining Λ with either the control gain K_R or K_V can also uniquely determine a series of guidance laws, as the other one can be analytically calculated. Since the explicit boundary of K_R can be derived from the stability conditions of the feedback system, the term Λ and K_R are learned in R-ZEM/ZEV. Then, the gain K_V can be analytically determined via Eqs. (18) and (19), from which the velocity increment in Eq. (16) can be obtained:

$$K_V = \frac{\Lambda^2 + 4K_R}{2\Lambda} - K_R - 1 \tag{20}$$

In RL, the exploration ranges of the learned parameters are $K_R \in [0, 12]$ and $\Lambda \in [2.1, 6.9]$, respectively. As this work focuses more on the robust trajectory design for the whole mission, the detailed derivations and analysis of the exploration range are not presented here, and can be referred to Ref. [28].

In the uncertain scenarios, the stochastic equations of motion of the EP spacecraft are identical to those in the ITPs, with the specific state propagations detailed in Eqs. (4) - (7). The state ($\boldsymbol{r}_{i,k}^{tgt}$, $\boldsymbol{v}_{i,k}^{tgt}$) in Eq. (14) can be obtained by back-propagating the target state ($\boldsymbol{r}_i^{tgt}$, $\boldsymbol{v}_i^{tgt}$) of the AP to the current step $t_{i,k}$ under Keplerian two-body dynamics. In this way, the agent state can be updated. Besides, at the final time t_f of the whole mission, i.e., the end step t_{L,N_L} of the last AP (with phase number $L = 2(n+1)$, corresponding to $i = L$), the spacecraft executes an instantaneous maneuver to match the velocity $\boldsymbol{v}_{tgt}$ of the rendezvous target, as in [2] [4]:

$$\Delta \boldsymbol{V}_{L,N_L} = min\left(\left\|\boldsymbol{v}_{tgt} - \boldsymbol{v}_{L,N_L}\right\|, \Delta V_{L,N_L}^{\max}\right)\frac{\boldsymbol{v}_{tgt} - \boldsymbol{v}_{L,N_L}}{\left\|\boldsymbol{v}_{tgt} - \boldsymbol{v}_{L,N_L}\right\|} \tag{21}$$

$$\begin{bmatrix} \boldsymbol{r}_{L,f} \\ \boldsymbol{v}_{L,f} \\ m_{L,f} \\ t_{L,f} \end{bmatrix} = \begin{bmatrix} \boldsymbol{r}_{L,N_L} \\ \boldsymbol{v}_{L,N_L} + \Delta \boldsymbol{V}_{L,N_L} \\ m_{L,N_L} \exp\left(-\left\|\Delta \boldsymbol{V}_{L,N_L}\right\| / u_{eff}\right) \\ t_{L,N_L} \end{bmatrix} \tag{22}$$

where the subscripts 'L' and 'f' indicate the last trajectory phase and the final time, respectively. It is worth noting that in the multi-phase strategy, no end velocity correction is performed on other phases with number $i < L$, so as to ensure smooth connection between the low-thrust trajectory phases. Thus, there will always be an undesired velocity deviation at the end of the ITPs, as the last part of the ITPs is usually non-ballistic, especially in uncertain scenarios. Moreover, since the end position accuracy of the RL-based robust trajectory of the ITP is only on the order of $10^{-3} \sim 10^{-4}$ AU (AU = 149597870.66 km) [2] [4], deviations in both position and velocity may render the target state of the AP unreachable for the EP spacecraft within the limited time. For this, a trade-off factor technique is proposed, and a reward function with reachability constraints is designed.

C. Reward Function Designing

In RL, the reward function is an important part of MDPs. Similarly, the reward functions are also designed separately for the ITPs and APs in this subsection. Firstly, to address the issue of failing to reach the target state of the AP due to terminal state deviations from the preceding ITP, a trade-off factor $\kappa \in [0, 1]$ between the optimality and robustness is defined to modify the original initial nominal state of the AP.

$$\begin{bmatrix} \bar{\boldsymbol{r}}_{\mathrm{AP},0} \\ \bar{\boldsymbol{v}}_{\mathrm{AP},0} \end{bmatrix} = \kappa \begin{bmatrix} \boldsymbol{r}_{\mathrm{AP},0} \\ \boldsymbol{v}_{\mathrm{AP},0} \end{bmatrix} + (1-\kappa) \begin{bmatrix} \hat{\boldsymbol{r}}_{\mathrm{AP},0} \\ \hat{\boldsymbol{v}}_{\mathrm{AP},0} \end{bmatrix} \tag{23}$$

where ($\bar{\boldsymbol{r}}_{\mathrm{AP},0}$, $\bar{\boldsymbol{v}}_{\mathrm{AP},0}$) is the modified initial nominal state of the AP, and ($\boldsymbol{r}_{\mathrm{AP},0}$, $\boldsymbol{v}_{\mathrm{AP},0}$) is the original initial nominal state of the AP taken from the fuel-optimal trajectory of the whole mission, which is solved by the indirect method [14]. The state ($\hat{\boldsymbol{r}}_{\mathrm{AP},0}$, $\hat{\boldsymbol{v}}_{\mathrm{AP},0}$) is the initial state that can reach the target state of the AP without control, which can be obtained by back-propagating the target state ($\boldsymbol{r}_i^{tgt}$, $\boldsymbol{v}_i^{tgt}$) of the AP to the initial time of the current AP. From Eq. (23), it can be seen that as the factor κ decreases, the modified ($\bar{\boldsymbol{r}}_{\mathrm{AP},0}$, $\bar{\boldsymbol{v}}_{\mathrm{AP},0}$) becomes closer to ($\hat{\boldsymbol{r}}_{\mathrm{AP},0}$, $\hat{\boldsymbol{v}}_{\mathrm{AP},0}$), indicating that the target state of the AP can be reached with a lower velocity increment. Thus, compared to the original ($\boldsymbol{r}_{\mathrm{AP},0}$, $\boldsymbol{v}_{\mathrm{AP},0}$), the EP spacecraft starting from the modified ($\bar{\boldsymbol{r}}_{\mathrm{AP},0}$, $\bar{\boldsymbol{v}}_{\mathrm{AP},0}$) has more control margins to compensate for uncertainties during the AP, especially in the cases where the engine is required to operate at maximum thrust throughout the AP in the original fuel-optimal solution. Besides, modifying this initial nominal state induces an alteration in the overall trajectory configuration of the mission, resulting in a reduction of the fuel optimality of the mission trajectory. This reduction becomes more dramatic with a smaller factor κ, hence κ is called the trade-off factor between optimality and robustness.

In particular, the EP spacecraft with states in the vicinity of the modified ($\bar{\boldsymbol{r}}_{\mathrm{AP},0}$, $\bar{\boldsymbol{v}}_{\mathrm{AP},0}$) can usually reach the target state of the AP in lower velocity increments. Thus, as the terminal target state ($\boldsymbol{r}_i^{tgt}$, $\boldsymbol{v}_i^{tgt}$) of the preceding

ITP, the modified ($\bar{\boldsymbol{r}}_{\mathrm{AP},0}$, $\bar{\boldsymbol{v}}_{\mathrm{AP},0}$) can better guide the agent to explore the robust policies of the ITP, so that the terminal states of the RL-based trajectories of the ITP can better satisfy the reachability constraints of the AP.

Compared with the existing method [33] that compensates for terminal deviations of ITPs by adjusting the GA or rendezvous time, the proposed trade-off factor technique is more suitable for the RL-based multi-phase robust trajectory design scheme. The reasons are as follows. In the multi-phase strategy, the terminal target state of an ITP needs to be pre-set to facilitate the RL training for robust policies of the ITPs. For the pre-set target state of the ITP before GA, the time that GA event can be altered is actually very limited due to the weak maneuverability of EP spacecraft and the short approaching period of a few tens of days. As analyzed in Sect. IV, small variations of the GA time within the limited range would lead to substantial additional fuel consumption owing to missing the optimal GA window. In the trade-off factor technique, the factor κ can be traversed by solving the fuel-optimal two-point boundary value problems (TPBVP) of the phases using the indirect method, and then a suitable κ can be rapidly determined. A brief flowchart is shown in Fig. 2, where the overall framework of the proposed method is presented. Besides, it is worth noting that this technique provides greater flexibility than the method in [33] [41], where a fixed-time pre-GA coast arc is enforced on the nominal trajectory to enhance robustness against uncertainties and facilitate navigation before GA [5]. This is because the fixed-time coast arc inherently corresponds to the case of $\kappa = 0$ in the trade-off factor technique. Particularly, the trade-off factor technique makes the multi-phase strategy also applicable to the cases with multiple GA maneuvers, as the coupling effects among different GA events under uncertainties can be significantly reduced.

1. Reward with Reachability constraint for Interplanetary Transfer

Now, the reward function of the ITPs is designed, where the reachability constraint of the target state of the AP is modeled analytically and incorporated into the reward. For the scenario *unp*, the reward $r_{i,k}$ is

$$r_{i,k} = -c_1\mu_{i,k} - c_2\Delta u_{i,k-1} - c_3 e_{i,k} + c_4 e^{-\lambda_1 e_{i,k}} + c_5(\Delta v_{i,k}^{orb} - \Delta v_{i,\max}^{orb}) + p_{i,k} + bonus \tag{24}$$

where

$$\mu_{i,k-1} = \begin{cases} m_{i,k-1} - m_{i,k} & \text{if } 1 \le k < N_i \\ m_{i,N_i-1} - m_{i,f} & \text{if } k = N_i \end{cases} \tag{25}$$

$$\Delta u_{i,k-1} = \max\left(0, \left\|\boldsymbol{u}_{i,k-1}\right\| - \Delta V_{i,k-1}^{\max}\right) \tag{26}$$

$$e_{i,k} = \begin{cases} 0 & \text{if } k < N_i \\ \max\left(\left\|\boldsymbol{r}_{i,f} - \boldsymbol{r}_i^{tgt}\right\| / \left\|\boldsymbol{r}_i^{tgt}\right\|, \left\|\boldsymbol{v}_{i,f} - \boldsymbol{v}_i^{tgt}\right\| / \left\|\boldsymbol{v}_i^{tgt}\right\|\right) & \text{if } k = N_i \end{cases} \tag{27}$$

$$\Delta v_{i,k}^{orb} = \begin{cases} 0 & \text{if } k < N_i \\ \left\|\boldsymbol{X}_{sc} - \hat{\boldsymbol{X}}_{\mathrm{AP},0}\right\| & \text{if } k = N_i \end{cases} \tag{28}$$

$$\boldsymbol{X}_{sc} = \left[\frac{\boldsymbol{r}_{i,N_i}}{tof_{\mathrm{AP}}} + \boldsymbol{v}_{i,N_i}; \frac{\boldsymbol{r}_{i,N_i}}{tof_{\mathrm{AP}}}; \frac{\boldsymbol{r}_{i,N_i}(tof_{\mathrm{AP}})}{tof_{\mathrm{AP}}} - \boldsymbol{v}_{i,N_i}(tof_{\mathrm{AP}}); \frac{\boldsymbol{r}_{i,N_i}(tof_{\mathrm{AP}})}{tof_{\mathrm{AP}}}\right] \in \mathbb{R}^{12} \tag{29}$$

$$\hat{\boldsymbol{X}}_{\mathrm{AP},0} = \left[\frac{\hat{\boldsymbol{r}}_{\mathrm{AP},0}}{tof_{\mathrm{AP}}} + \hat{\boldsymbol{v}}_{\mathrm{AP},0}; \frac{\hat{\boldsymbol{r}}_{\mathrm{AP},0}}{tof_{\mathrm{AP}}}; \frac{\hat{\boldsymbol{r}}_{\mathrm{AP},0}(tof_{\mathrm{AP}})}{tof_{\mathrm{AP}}} - \hat{\boldsymbol{v}}_{\mathrm{AP},0}; \frac{\hat{\boldsymbol{r}}_{\mathrm{AP},0}(tof_{\mathrm{AP}})}{tof_{\mathrm{AP}}}\right] \in \mathbb{R}^{12} \tag{30}$$

$$\boldsymbol{r}_{ast,dep} = \boldsymbol{r}_{i,N_i} + \boldsymbol{v}_{i,N_i} t, \quad \boldsymbol{r}_{ast,arr} = \hat{\boldsymbol{r}}_{\mathrm{AP},0} + \hat{\boldsymbol{v}}_{\mathrm{AP},0} t, \quad \boldsymbol{r}_s = \boldsymbol{r}_{i,N_i} + \boldsymbol{v}_{s,0} t \tag{31}$$

$$\boldsymbol{v}_{s,0} = \frac{\hat{\boldsymbol{r}}_{\mathrm{AP},0} - \boldsymbol{r}_{i,N_i}}{tof_{\mathrm{AP}}} + \hat{\boldsymbol{v}}_{\mathrm{AP},0} \tag{32}$$

$$p_{i,k} = \begin{cases} -2 & \text{if } \left\|\boldsymbol{r}_{i,k}\right\| < F_1 \text{ or } \left\|\boldsymbol{r}_{i,k}\right\| > F_2 \text{ or } \Delta u_{i,k} > 0 \\ 0 & \text{otherwise} \end{cases} \tag{33}$$

$$\Delta v_{i,\max}^{orb} = \eta_i T_{\max} tof_{\mathrm{AP}} / m_{i,N_i} \tag{34}$$

$$bonus = \begin{cases} 5 & \text{if } \Delta v_{i,N_i}^{orb} \le \Delta v_{i,\max}^{orb} \\ 0 & \text{otherwise} \end{cases} \tag{35}$$

In Eq. (24), $\mu_{i,k}$ is the fuel cost term, and the negative coefficient is to motivate the agent to save propellant. The term $\Delta u_{i,k}$ denotes the thrust magnitude constraint of the velocity increment $\boldsymbol{u}_{i,k}$: $\left\|\boldsymbol{u}_{i,k-1}\right\| \le \Delta \boldsymbol{V}_{i,k-1}^{\max}$. In the case of $\left\|\boldsymbol{u}_{i,k-1}\right\| > \Delta \boldsymbol{V}_{i,k-1}^{\max}$ during RL training, a normalization procedure is applied via the saturation function in Eq. (16), which is beneficial for the agent's exploration of bang-bang control profiles. For the reward related to the end-state deviations $e_{i,k}$, an exponential term is added to make the scalar reward more sensitive to $e_{i,k}$ [4]. Besides, a virtual velocity correction in Eqs. (21) and (22) is assumed to simplify the design of the reward, and the resulting final virtual state ($\boldsymbol{r}_{i,f}$, $\boldsymbol{v}_{i,f}$, $m_{i,f}$) is limited to the computation of $e_{i,k}$ and fuel consumption of the virtual correction. In our experiment, such reward with virtual correction is more conducive to the exploration of the optimal policies. In particular, the orbital indicator $\Delta v_{i,k}^{orb}$ is adopted to quantify the reachability of the target state the AP from the end states of the preceding ITP and is incorporated into the reward. The indicator $\Delta v_{i,k}^{orb}$ is originally developed to quickly estimate the cost of hops between two asteroids, assuming that the asteroids and spacecraft undergo uniform rectilinear motion due to the short-arc transfer characteristic [42]. This assumption is also applicable to the APs, where ($\boldsymbol{r}_{i,N_i}$, $\boldsymbol{v}_{i,N_i}$) and ($\hat{\boldsymbol{r}}_{\mathrm{AP},0}$, $\hat{\boldsymbol{v}}_{\mathrm{AP},0}$) can be regarded as the initial states of the two asteroids. The states of the asteroids and spacecraft are updated based on Eq. (31) in this assumed model, in which $\boldsymbol{r}_{ast,\,dep}$, $\boldsymbol{r}_{ast,\,arr}$, and $\boldsymbol{r}_s$ denote the real-time position of the departure asteroid, arrival asteroid, and spacecraft, respectively, and $\boldsymbol{v}_{s,\,0}$ is the spacecraft's initial velocity. By matching the position and velocity of the spacecraft at the time tof_{AP} (the duration of the AP) with those of the arrival asteroid, the initial velocity $\boldsymbol{v}_{s,\,0}$ can be derived, as in Eq. (32). Then, the corresponding velocity increments can be computed, i.e., the differences between the first two terms of $\boldsymbol{X}_{sc}$ and $\hat{\boldsymbol{X}}_{\mathrm{AP},0}$. The last two terms in $\boldsymbol{X}_{sc}$ and $\hat{\boldsymbol{X}}_{\mathrm{AP},0}$ are incorporated to correct the oversimplified linear dynamics by looking at the differences of the states at the end time of the AP [42], where ($\boldsymbol{r}_{i,N_i}(tof_{\mathrm{AP}})$, $\boldsymbol{v}_{i,N_i}(tof_{\mathrm{AP}})$) and ($\hat{\boldsymbol{r}}_{\mathrm{AP},0}(tof_{\mathrm{AP}})$, $\hat{\boldsymbol{v}}_{\mathrm{AP},0}(tof_{\mathrm{AP}})$) are the actual states of the two asteroids at the time tof_{AP} under two-body dynamics. The item $p_{i,k}$ is a path constraint to facilitate the agent to quickly reach a feasible solution range, where the parameters F_1 and F_2 are determined by the lower and upper bounds of the distance of the EP spacecraft from the Sun during the ITP. The term *bonus* acts as a significant incentive when the reachability constraint is satisfied, where $\Delta v_{i,\max}^{orb}$ is the maximum velocity increment that can be converted into a low-thrust trajectory in the AP to reach the target state. The values of -2 in $p_{i,k}$ and 5 in *bonus* are taken to provide a direct signal regarding the quality of the current policy to the training agent, while avoiding excessive changes in reward value that are not conducive to the learning of the value function in RL.

This setting facilitates the agent to explore the optimal policy. The term η_i is a discount factor for the conversion from the maximum increment that can be delivered by the EP engine in the AP to the corresponding low-thrust trajectory, which is an empirical value and determined experimentally. The relevant coefficients are also determined experimentally and will be given in Sect. IV, which is also true for the following rewards.

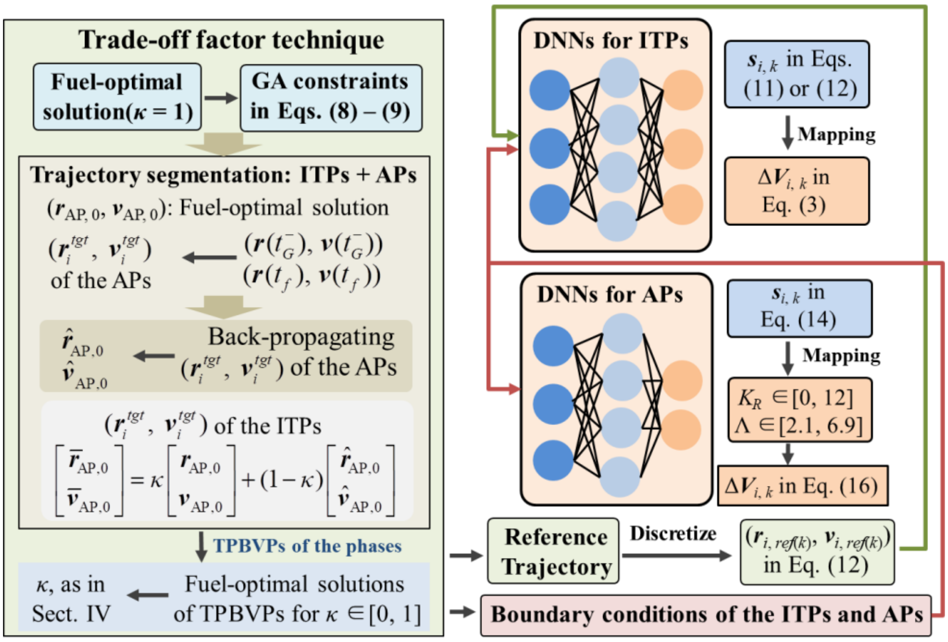


Fig. 2 Overall framework of the RL-based multi-phase robust low-thrust GA trajectory design method

For the uncertain scenarios, the reward $r_{i,k}$ is

$$r_{i,k} = -c_1\mu_{i,k} - c_2\Delta u_{i,k-1} + c_3(\Delta v_{i,k}^{orb} - \Delta v_{i,\max}^{orb}) + \zeta(\boldsymbol{es}_{i,k}) + \xi(e_{i,k}) + bonus \tag{36}$$

where

$$\boldsymbol{es}_{i,k} = \left[er_{i,k},\ ev_{i,k}\right] = \left[\left\|\boldsymbol{r}_{i,k} - \boldsymbol{r}_{i,ref(k)}\right\| / \left\|\boldsymbol{r}_{i,ref(k)}\right\|,\ \left\|\boldsymbol{v}_{i,k} - \boldsymbol{v}_{i,ref(k)}\right\| / \left\|\boldsymbol{v}_{i,ref(k)}\right\|\right] \tag{37}$$

$$\zeta(\boldsymbol{es}_{i,k}) = \frac{k}{N_i}\left(-c_4 er_{i,k} - c_5 ev_{i,k} + c_6 e^{-\lambda_1 \cdot er_{i,k}} + c_7 e^{-\lambda_1 \cdot ev_{i,k}}\right) \tag{38}$$

$$\xi(e_{i,k}) = -c_8 e_{i,k} + c_9 e^{-\lambda_2 e_{i,k}} \tag{39}$$

In Eq. (36), the terms $\mu_{i,k}$, $\Delta u_{i,k}$, $\Delta v_{i,k}^{orb}$, $\Delta v_{i,\max}^{orb}$, and *bonus* are the same as those in Eq. (24). The term $\boldsymbol{es}_{i,k}$ denotes the deviations of the actual states of the EP spacecraft under uncertainties from the reference ones, thus, $\zeta(\boldsymbol{es}_{i,k})$ is to drive the spacecraft's motion within the vicinity of the reference trajectory. As well, the increasing coefficients over time are intended to prevent sub-optimality due to over-tracking the reference trajectory.

2. *Reward for Approaching Phases*

For the robust guidance of the APs, the reward function is

$$r_{i,k} = -c_1\mu_{i,k} - c_2 e_{i,k} + c_3 e^{-\lambda_1 e_{i,k}} + bonus \tag{40}$$

where

$$e_{i,k} = \begin{cases} 0 & \text{if } k < N_i \\ \max\left(\left\| \boldsymbol{r}_{i,N_i} - \boldsymbol{r}_i^{tgt} \right\| / L_{f,AP},\ \left\| \boldsymbol{v}_{i,N_i} - \boldsymbol{v}_i^{tgt} \right\| / V_{f,AP} \right) & \text{if } k = N_i \end{cases} \tag{41}$$

$$bonus = \begin{cases} 5 & \text{if } e_{i,N_i} < \varepsilon \\ 0 & \text{otherwise} \end{cases} \tag{42}$$

From Eq. (40), it can be seen that the end state deviation $e_{i,k}$ of the APs is slightly different from that of the ITPs, in which $L_{f,\text{AP}} = 1\times10^4$ km, $V_{f,\text{AP}} = 10$ km/s. This dimensionless of the end deviations is due to the fact that the expected end accuracy is kilometer-level in position and meters per second in velocity, whereas the normalized accuracy in RL training is typically up to 10^{-4} for the investigated problem, as in [4] [28]. In the term *bonus*, the terminal tolerance ε is set to 0.001.

D. Proximal Policy Optimization Algorithm

The state-of-the-art RL algorithm, Proximal Policy Optimization (PPO) [26], is used to train deep neural networks (DNNs) for robust policies. PPO is a model-free policy-gradient training algorithm that has achieved promising results in many astrodynamics problems. The adopted basic architecture of PPO is the Actor-Critic (AC) framework, where the Actor and the Critic run in parallel and update concurrently during the training.

In the AC schematic, the Actor samples the current parameterized policy $\pi_\Theta(\boldsymbol{a}_k \mid \boldsymbol{s}_k)$ and updates the DNN's parameters Θ based on policy gradient. As the basics of PPO are the same for all trajectory phases, the subscript i used to indicate the phases is not shown in this subsection. The Critic returns a parameterized state-value function $V^{\pi_\Theta}(\boldsymbol{s}_k)$ to evaluate the Actor performance. In PPO, the advantage function $A^{\pi_\theta}(\boldsymbol{s}_k, \boldsymbol{a}_k)$ is usually adopted to compute the expected gradient of the merit index $J(\Theta)$ in RL due to its lower variance [26]:

$$J(\Theta)=\left(\mathop{\mathbb{E}}_{\tau\sim\pi_\Theta}\left[\sum_{k=0}^{N-1}\gamma^k r_k\right]\right) \tag{43}$$

$$V^{\pi_\Theta}\left(\boldsymbol{s}_k\right)=\mathop{\mathbb{E}}_{r-\pi_\Theta}\left[\sum_{l=0}^{N-k}\gamma^l r_{k+l}\mid \boldsymbol{s}_k\right] \tag{44}$$

$$Q^{\pi_\Theta}\left(\boldsymbol{s}_k,\boldsymbol{a}_k\right)=\mathop{\mathbb{E}}_{\tau\sim\pi_\Theta}\left[\sum_{l=0}^{N-k}\gamma^l r_{k+l}\mid \boldsymbol{s}_k,\boldsymbol{a}_k\right] \tag{45}$$

$$A^{\pi_\Theta}\left(\boldsymbol{s}_k,\boldsymbol{a}_k\right)=Q^{\pi_\Theta}\left(\boldsymbol{s}_k,\boldsymbol{a}_k\right)-V^{\pi_\Theta}\left(\boldsymbol{s}_k\right) \tag{46}$$

where γ is the discount factor in $J(\Theta)$, and $Q^{\pi_\Theta}\left(\boldsymbol{s}_k,\boldsymbol{a}_k\right)$ is the action-value function, which together with $V^{\pi_\Theta}(\boldsymbol{s}_k)$ are two fundamental functions in RL. The advantage function $A^{\pi_\Theta}(\boldsymbol{s}_k,\boldsymbol{a}_k)$ is used to measure the increase in reward due to a specified action compared to the default policy. In practice, $A^{\pi_\Theta}\left(\boldsymbol{s}_k,\boldsymbol{a}_k\right)$ is approximated by a so-called generalized advantage estimation $\hat{A}_k$ for the convenience of computation [43]:

$$\hat{A}_k=\sum_{l=0}^{N-1}(\gamma\lambda)^l\vartheta_{k+l}^V \qquad \vartheta_k^V=r_k+\gamma V^{\pi_\theta}\left(\boldsymbol{s}_{k+1}\right)-V^{\pi_\theta}\left(\boldsymbol{s}_k\right) \tag{47}$$

where ϑ_k^V is the temporal-difference residual of $V^{\pi_\Theta}(\boldsymbol{s}_k)$ with the discount γ and λ.

In PPO, a clipping mechanism is utilized to avoid too large policy updates during the training process, and the resulting clipped objective function $J_{clip}(\Theta)$ is [26]

$$J_{clip}(\Theta)=\mathop{\mathbb{E}}_{\tau\sim\pi_\Theta}\left[\frac{1}{N}\sum_{k=0}^{N-1}\min\left\{\left(\hat{r}_k(\Theta),clip\left(\hat{r}_k(\Theta),1-э,1+э\right)\right)\hat{A}^\pi(\boldsymbol{s}_k,\boldsymbol{a}_k)\right\}\right] \tag{48}$$

$$\hat{r}_k(\Theta)=\pi_\Theta\left(\boldsymbol{a}_k\mid \boldsymbol{s}_k\right)/\pi_{\Theta old}\left(\boldsymbol{a}_k\mid \boldsymbol{s}_k\right) \tag{49}$$

where э is the clipping factor in the *clip* function, which can be consulted in [26]. The ratio $\hat{r}_k(\Theta)$ is to quantify the differences between the new and old policies. From Eq. (48), it can be seen that this surrogate function limits the agent to update the policies in a small pace, thus improving the stability of the training.

In addition, two additional terms, a mean-squared value function error $H(\Theta)$ of the $V^{\pi_\Theta}(\boldsymbol{s}_k)$ and an entropy term $S(\Theta)$ for ensuring sufficient exploration, are usually appended to the objective [26]:

$$J^{\mathrm{PPO}}(\Theta)=J_{clip}(\Theta)-coe_1H(\Theta)+coe_2S(\Theta) \tag{50}$$

$$H(\Theta)=\mathop{\mathbb{E}}_{\tau\sim\pi_\Theta}\left[\frac{1}{N}\sum_{k=0}^{N-1}\frac{1}{2}\left(V_\Theta\left(\boldsymbol{s}_k\right)-\sum_{l=k}^{N-1}\gamma^{l-k}r_l\right)^2\right] \tag{51}$$

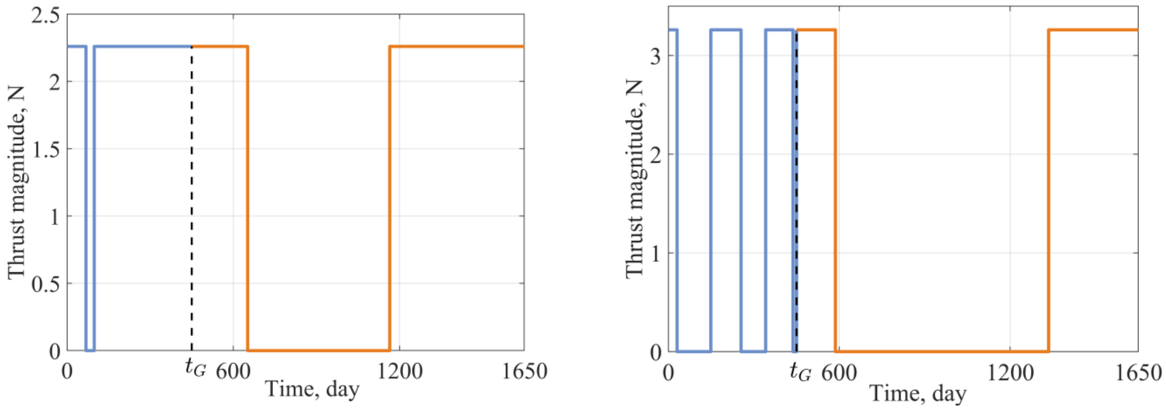
$$S(\Theta) = \mathop{\mathbb{E}}_{\tau \sim \pi_\Theta} \left[\frac{1}{N} \sum_{t=0}^{N-1} \mathbb{E}_{a \sim \pi_\Theta(\cdot|s_k)} \left[-\log \pi_\Theta \left(\boldsymbol{a} \,|\, \boldsymbol{s}_k \right) \right] \right] \tag{52}$$

where coe_1 and coe_2 are the relevant coefficients of the terms, and $J^{\mathrm{PPO}}(\Theta)$ is the final objective in PPO.

For the implementation of PPO, the open-source RL library Stable-Baseline3 is utilized [44]. The networks of the Actor and Critic are both multilayer perceptron (MLP), which consist of an input layer, hidden layers and an output layer, and the architectures of DNNs are heuristically designed, as in [2] [12]. The number of neurons n_{in} of input layer for the Actor and Critic is the same as the state dimension of the corresponding MDP. For the output layer of the Actor, the number of neurons n_{out} is 3 and 2 for the ITPs and APs, respectively, and $n_{out} = 1$ for the Critic of both the ITPs and APs. The hidden layers are composed of three layers of size $h_1 = 10n_{in}$, $h_2 = \sqrt{h_1 h_3}$ and $h_3=10n_{out}$. The DNN architectures for the ITPs and APs are the only modification made in the application of this RL library. Besides, the experimental platform is a computer assembled with an Intel Core i5-10600KF CPU @ 4.10GHz and a NVIDIA GeForce GTX 1660 SUPER GPU.

IV. Numerical Results and Discussion

To validate the performance of the proposed RL-based multi-phase robust low-thrust GA trajectory design method, a three-dimensional Earth-Earth-Jupiter mission is considered. The mission parameters are summarized in Table 1, which are taken from Refs. [8] [9]. The maximum thrust T_{max} is slightly enlarged than that in [8] [9], and the corresponding specific impulse I_{sp} is calculated by assuming the same engine power as that in [8] [9]. This is because the EP engine needs to operate at maximum thrust for almost the entire trajectory before GA under the thrust parameter in [8] [9], indicating that the mission trajectory is highly-sensitive to uncertainties. If robust trajectory design is attempted directly, the convergence rates for satisfying the constraints of GA and terminal rendezvous would be unacceptably low. Appropriately increasing the thrust magnitude can enhance the duration of coasting arcs along the trajectory, thus decreasing the sensitivity of the reference trajectory to uncertainties. This is essentially similar to the practice in [33] [41], where the nominal trajectory is modified prior to robust trajectory design by incorporating a forced pre-GA coast arc. Nevertheless, the simulation results can still demonstrate the proposed method. In Fig. 3, the thrust magnitudes versus time under the original thrust in [8] [9] and the adopted thrust in this study are depicted for better illustration.



(a) T_{max} = 2.26 N in Refs . [8] [9]   (b) T_{max} = 3.26 N in this study

Fig. 3 Thrust profile over the Earth-Earth-Jupiter mission under different thrust parameters

In RL, the agent states are nondimensionalized by scaling factors. For the ITPs, scaling factors include the astronomical unit AU = 149597870.66 km, corresponding circular velocity VU = 29.7847 km/s, and initial mass m_0. As for the APs, the normalization of the states related to position and velocity is conducted separately on three axes, enabling the DNN to better capture the differences in states across the steps, that is

$$\begin{cases} Lx_i = \left| x_{i,0} - x_{i,0}^{tgt} \right| + 3\sigma_{x,i}, \; Ly_i = \left| y_{i,0} - y_{i,0}^{tgt} \right| + 3\sigma_{y,i}, \; Lz_i = \left| z_{i,0} - z_{i,0}^{tgt} \right| + 3\sigma_{z,i} \\ Vx_i = \left| Vx_{i,0} - Vx_{i,0}^{tgt} \right| + 3\sigma_{Vx,i}, \; Vy_i = \left| Vy_{i,0} - Vy_{i,0}^{tgt} \right| + 3\sigma_{Vy,i}, \; Vz_i = \left| Vz_{i,0} - Vz_{i,0}^{tgt} \right| + 3\sigma_{Vz,i} \end{cases} \tag{53}$$

where (Lx_i, Ly_i, Lz_i) and (Vx_i, Vy_i, Vz_i) are the three-axis scaling factors for the position and velocity of the AP, respectively. The terms ($x_{i,0}$, $y_{i,0}$, $z_{i,0}$) and ($Vx_{i,0}$, $Vy_{i,0}$, $Vz_{i,0}$) are the three-axis components of the initial state $\boldsymbol{r}_{i,0}$ and $\boldsymbol{v}_{i,0}$ of the spacecraft in the AP, and ($x_{i,0}^{tgt}$, $y_{i,0}^{tgt}$, $z_{i,0}^{tgt}$) as well as ($Vx_{i,0}^{tgt}$, $Vy_{i,0}^{tgt}$, $Vz_{i,0}^{tgt}$) are the components of $\boldsymbol{r}_{i,0}^{tgt}$ and $\boldsymbol{v}_{i,0}^{tgt}$. It is worth mentioning that ($\sigma_{x,i}$, $\sigma_{y,i}$, $\sigma_{z,i}$) and ($\sigma_{Vx,i}$, $\sigma_{Vy,i}$, $\sigma_{Vz,i}$) are the standard deviations of the end position and velocity from the preceding ITP, which are obtained from the Monte-Carlo (MC) simulations of the robust policies in the ITP. In the multi-phase strategy, the robust policies of each phase are trained sequentially, where initial state deviations are obtained from the MC simulations of the trained policies in the preceding phase. This manner ensures smooth connection between the phases during training.

Table 1 Problem parameters for the Earth-Earth-Juipter mission

Parameters	Value
Initial time t_0, TDB	29. Sep. 2015
GA time t_G, TDB	13. Dec. 2016; (450 days)
Flight time t_f, day	1650
Initial position of the Earth $\boldsymbol{r}_E(t_0)$, km	[149237771.6545, 13907019.4862, -15826.8605]
Initial velocity of the Earth $\boldsymbol{v}_E(t_0)$, km/s	[-3.2452, 29.5363, -0.0312]
Launch v_∞ on Earth, km/s	0.71
Final position $\boldsymbol{r}_{tgt}$, km	[183153189.0986, -754925706.4798, -162321.4695]
Final velocity $\boldsymbol{v}_{tgt}$, km/s	[12.5529, 3.7006, -0.3001]
Earth Gm μ_E, km^3/s^2	398600.435436
Minimum pericenter radius $r_{p,\min}$, km	6878.145
Initial mass m_0, kg	19820
T_{max}, N	3.26
I_{sp}, s	4160

Besides, the overall trajectory is divided into four phases ($2(n+1)$), as only a GA maneuver (i.e., n = 1) is included in this example, with ITPs of 400 and 1170 days before and after GA, and APs of 50 and 30 days prior to GA maneuver and terminal rendezvous, respectively. The number of discrete steps for the phases is $N_{1\sim4}$ = 50, 30, 100, 30. The PPO hyperparameters (Hyp) are given in Table 2, where n_t indicates the current training process and n^* is the total training steps. The parameters n_{env} and n_{step} are the number of independent realizations of the environment (i.e., parallel) and the number of sampling steps per Actor during training of a specific robust policy. The symbol α denotes the learning rate of DNNs, and n_{opt} is the number of epochs per update. These parameters are determined based on existing studies [2] [4] [28] and our experiments.

Table 2 Hyperparameters of Proximal Policy Optimization

Parameters	Value
γ	0.9999
λ	0.99
α	$2.5\times10^{-4}(1-n_t/n^*)$
$\ni$	$0.25(1-n_t/n^*)$
β_1	0.5
β_2	4.75×10^{-8}
n_{opt}	30
n_{env}	64
n_{step}	N_i

A. Deterministic gravity-assist trajectory optimization

Firstly, the performance of the proposed multi-phase trajectory design method is validated in the scenario *unp*. The reward coefficients for the two ITPs are presented in Table 3, where ITP_1 and ITP_2 correspond to the ITPs before and after GA. Similarly, these coefficients are initialized with reference to our previous work [4] [28], which are further refined according to the performance of the trained agent in our experiment. The values of parameters F_1 and F_2 are based on the fact that the distance of the spacecraft from the Sun during ITP_1 and ITP_2 must be within the respective intervals of [0.8 AU, 1.2 AU] and [0.95AU, 5.3 AU] in the simulated mission. The values of the discounted factor η_i are discussed in the next subsection. As seen, the reachability index coefficient c_5 is set to be two-stage, taking the value of 4 for the first 1/10 part of RL training and 10 for the rest. The stage node of RL training we adopted is an empirical value based on real-time terminal state errors in Eq. (27) in our experiment. This setting is drawn from [2], and it is indeed beneficial for training. As for the two APs, the reward coefficients are $c_1 = 10$, $c_2 = 20$, $c_3 = 1$, and $\lambda_1 = 1000$.

The trade-off factors κ for the two APs are separately traversed over the range [0, 1] with a step size of 0.05, where the phased fuel-optimal trajectories are solved by the indirect method [14]. For the AP (labeled as AP_1) before GA, $\kappa_1 = 0.6$, which is determined by restricting the fuel consumption corresponding to the modified ($\bar{\boldsymbol{r}}_{AP,0}$, $\bar{\boldsymbol{v}}_{AP,0}$) to be no more than 5% of the original optimal solution from time t_0 to GA time t_G. Then, for the AP prior to terminal rendezvous (AP_2), the total propellant consumption Δm_{total} for the whole mission is 4777.105 kg when its $\kappa_2 = 1$ and 4791.580 kg if $\kappa_2 = 0$. The difference is small relative to the total consumption, thus, we set κ_2 to 0.5 ($\Delta m_{total} = 4781.045$ kg). Besides, for the strategy of altering the GA time, the maximum allowable delay of GA time is only 2 days for the original initial state ($\boldsymbol{r}_{AP,0}$, $\boldsymbol{v}_{AP,0}$) in the AP_1, and the resulting Δm_{total} is 4840.704 kg. It can be seen that more propellant is consumed due to missing the optimal GA window in this example. Although it is possible to save fuel by further extending the final rendezvous time, this would result in more complexities in trajectory design or even mission failure for cases with multiple gravity assists. On the contrary, the proposed trade-off factors technique can avoid this drawback.

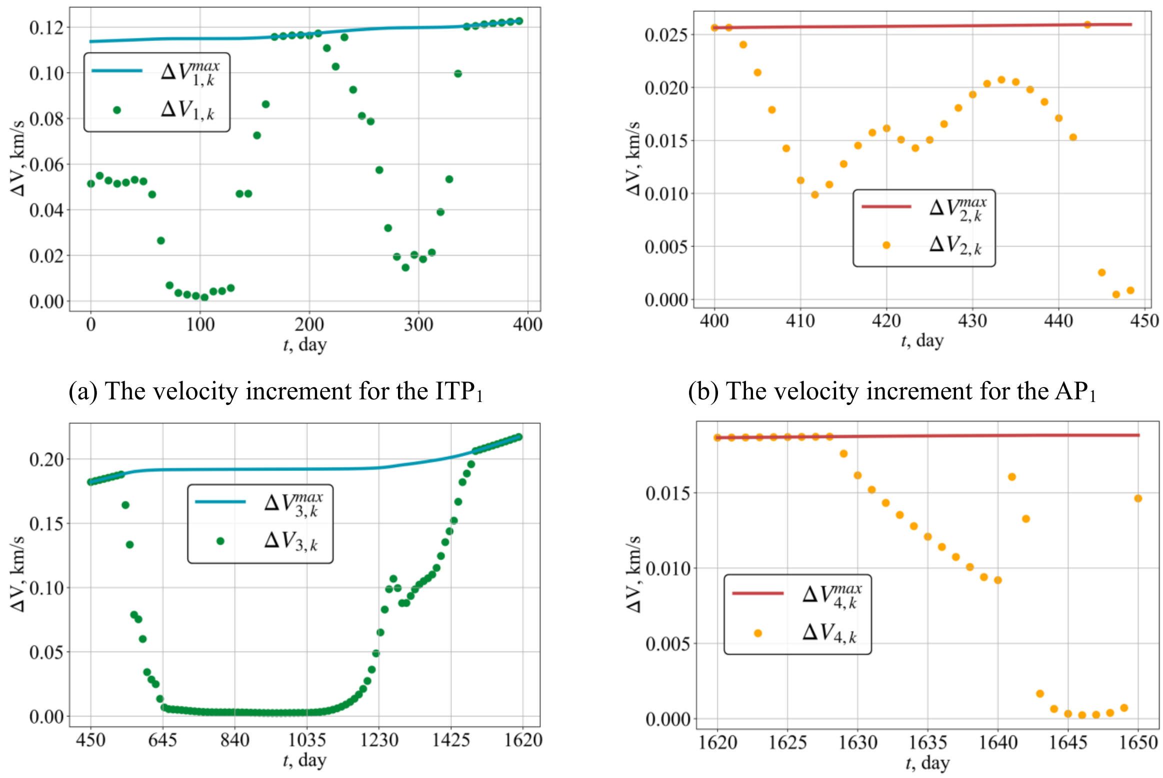
Table 3 Reward coefficients of the interplanetary transfer in the unperturbed scenario

Coefficient	c_1	c_2	c_3	c_4	c_5	λ_1	F_1	F_2	η_i
ITP_1	30	5	10	1.5	4, 10	1000	0.8	1.2	0.62
ITP_2	40	5	10	1.5	4, 10	1000	0.95	5.3	0.67

The relevant data of the deterministic optimal trajectory obtained from the proposed method are presented as the first data item in the tables for numerical results of robust trajectories, where ($\pi_{\text{ITP}_1}^{unp}$, $\pi_{\text{AP}_1}^{unp}$, $\pi_{\text{ITP}_2}^{unp}$, $\pi_{\text{AP}_2}^{unp}$) denote the trained optimal policies for the four trajectory phases in the scenario *unp*. The subscripts '$\text{ITP}_{1\sim2}$' and '$\text{AP}_{1\sim2}$' indicate the trajectory phases, and the superscript '*unp*' refers to the corresponding scenario, which is also true for the robust policies in these tables. The terms $\Delta r_{i,N_i}$ and $\Delta v_{i,N_i}$ (i = 1~3) denote the deviations of position and velocity at the end time of the respective phases, and $\Delta r_{4,f}$ and $\Delta v_{4,f}$ are the deviations at the final time of the mission. As well, the letter M of the 'Training steps' stands for 10^6.

From the data, it can be seen that the position errors at times t_G and t_f are less than 0.1 km, which is much lower than those of 10^{-3} ~ 10^{-4} AU reported in existing RL methods for interplanetary rendezvous [2] [4]. The total fuel consumption Δm_{total} is 4862.334 kg in the proposed method, which is 1.70% more than that of the phased fuel-optimal solutions corresponding to the modified ($\bar{\boldsymbol{r}}_{\text{AP},0}, \bar{\boldsymbol{v}}_{\text{AP},0}$) solved by the indirect method [14]. In the phased fuel-optimal trajectories, the masses (m_{i,N_i}) at the end of the four phases are 18300.358 kg, 18146.313 kg, 15142.137 kg, and 15038.955 kg, respectively. Comparing the propellant consumption of each phase between the two methods, it can be seen that the differences are mainly in AP_1 and AP_2. On the one hand, it is due to the fact that the robust guidance laws in the APs are inherently suboptimal when the gravitational acceleration is neither constant nor time-dependent, and it is derived from an energy-optimal index. On the other hand, additional propellant is consumed in the APs to compensate for the terminal state deviations of the ITPs. Nevertheless, the minor difference in Δm_{total} signifies the effectiveness of the proposed method.

The velocity increment magnitudes along the low-thrust trajectory in scenario *unp* are plotted in Fig. 4, where the increments $\Delta V_{1\sim4,\,k}$ for each phase are given separately for better illustration. As shown in Fig. 4, the velocity increments exhibit an approximate bang-bang control profile, especially for the ITPs, which is the characteristic of the fuel-optimal trajectory. Meanwhile, it is seen that the EP engine needs to operate at maximum thrust for tens (ITP$_1$) or even hundreds of days (ITP$_2$) immediately prior to the end of the ITPs, indicating that the EP spacecraft is highly sensitive to uncertainties during this trajectory segment. This is also the reason why a large number of training steps is required for the robust policies of the ITPs.



(a) The velocity increment for the ITP$_1$

(b) The velocity increment for the AP$_1$

(c) The velocity increment for the ITP$_2$

(d) The velocity increment for the AP$_2$

Fig. 4 The velocity increment $\Delta V_{i,\,k}$ over the mission time in scenario *unp*

B. Robust trajectory design in uncertain scenarios

Next, the performance of the proposed method is analyzed in the considered uncertain scenarios. The values of the uncertainty parameters in Sect. II. A are listed in Table 4, which are drawn from some existing studies on autonomous space missions [2] [3] [45]. The coefficients of the rewards of the ITPs for the uncertain scenarios are given in Table 5. In this study, the discount factors η_i are chosen conservatively in RL training, since the smaller reachability indices ($\Delta v_{1,N_1}^{orb}$, $\Delta v_{3,N_3}^{orb}$) typically indicate lower propellant consumption in the APs.

Thus, the suboptimal guidance laws in the APs will have less effect on fuel performance, as mentioned earlier. The larger η_i for ITP_2 is due to the fact the spacecraft is farther from the Sun and is with reduced mass in ITP_2 in this mission, especially in the vicinity of Jupiter, enabling enhanced orbital maneuverability of the spacecraft. As for the rewards of the APs, they are the same as those in the scenario *unp*.

Table 4 Uncertainty parameters in the mission

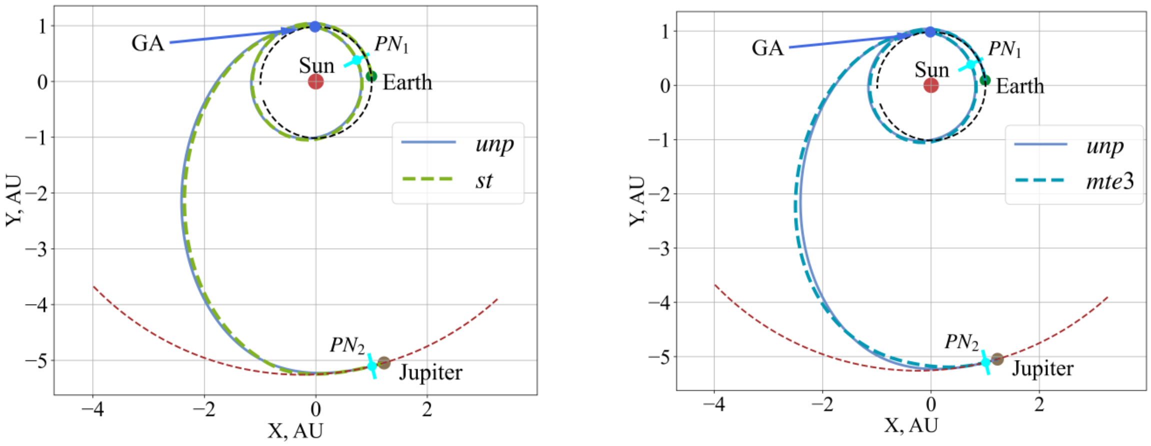
Parameters	$\sigma_{i,\,rs}$, km	$\sigma_{i,\,vs}$, m/s	$\sigma_{i,\,ro}$, km	$\sigma_{i,\,vo}$, m/s	$\sigma_{i,\,u}$	$\sigma_{i,\,Eu}$, deg	$N_{i,\,mte}$	$P_{i,\,mte}$
ITP	1.0	1.0	1.0	10.0	0.03	0.3	3	0.9
AP	0.1	0.1	0.1	0.1	0.01	0.1	3	0.9

Table 5 Reward coefficients of the interplanetary transfer in the uncertain scenarios

Coefficient	c_1	c_2	c_3	c_4	c_5	c_6	c_7	c_8	c_9	λ_1	λ_2	η_i
ITP_1	20	10	4, 10	0.2	0.02	0.2	0.02	10	1.5	500	1000	0.62
ITP_2	25	10	4, 10	0.2	0.02	0.6	0.06	10	1.5	500	1000	0.67

With the trained robust policies, the robust trajectory of each uncertain scenario can be obtained by running the robust policies in their respective unperturbed situations, as depicted in Fig. 5. In Fig. 5, two representative scenarios, *st* and *mte*3, are shown, where PN_1 (patched node 1) and PN_2 denote the starting positions of AP_1 and AP_2. The numerical results are given in Table 6 - Table 9, where the data of the four phases are presented separately. In Table 6 and Table 8, it can be seen that both the reachability indices $\Delta v_{1,N_1}^{orb}$ and $\Delta v_{3,N_3}^{orb}$ are below the prescribed maximum values in training ($\Delta v_{1,\max}^{orb}$ = 0.477 km/s, $\Delta v_{3,\max}^{orb}$ = 0.374 km/s), which can be calculated with the factor η_i in Table 5. In scenarios *st*, *obs*, and *ctr*, the final velocity errors $\Delta v_{4,f}$ are zero as there is an algebraically calculated velocity correction at the end of the last phase, and the magnitude of the correction is lower than the maximum admissible $\Delta V_{4,\,N_4}^{\max}$, as in Eq. (21).

Similar to the scenario *unp*, the position errors at time t_G among the robust trajectories are only on the order of kilometers, which is crucial for achieving the desired GA maneuver. The same is also true for the terminal state errors for final rendezvous. Compared with the final mass in *unp*, less than 3.0% propellant is sacrificed for robustness to the considered uncertainties, with a maximum of 2.98% in scenario *mte*3. As well, 6.06% and 4.73% more propellant is consumed along the robust trajectory in *mte*3, when comparing with the original fuel-optimal solution corresponding to ($\boldsymbol{r}_{\mathrm{AP},0}$, $\boldsymbol{v}_{\mathrm{AP},0}$) and the phased fuel-optimal solution of the modified ($\bar{\boldsymbol{r}}_{\mathrm{AP},0}$, $\bar{\boldsymbol{v}}_{\mathrm{AP},0}$). The reasons for these differences have been explained in the preceding subsection. Besides, the training steps of ITP_2 are obviously more than those of ITP_1, due to the longer flight time and more discrete steps in ITP_2.



(a) Robust trajectory of the scenario *st* (b) Robust trajectory of the scenario *mte*3

Fig. 5 Robust trajectories of the considered uncertainties

Table 6 Numerical results of robust trajectories of ITP_1

Policy	Training steps	Results			
		m_{1,N_1} , kg	$\Delta r_{1,N_1}$, AU	$\Delta v_{1,N_1}$, VU	$\Delta v^{orb}_{1,N_1}$, km/s
$\pi^{unp}_{\mathrm{ITP}_1}$	150M	18310.850	2.931×10^{-3}	2.910×10^{-3}	0.453
$\pi^{st}_{\mathrm{ITP}_1}$	150M	18224.367	1.487×10^{-3}	3.036×10^{-3}	0.399
$\pi^{obs}_{\mathrm{ITP}_1}$	100M	18265.649	2.525×10^{-3}	2.378×10^{-3}	0.463
$\pi^{ctr}_{\mathrm{ITP}_1}$	150M	18210.172	1.124×10^{-3}	8.759×10^{-4}	0.392
$\pi^{mte1}_{\mathrm{ITP}_1}$	150M	18199.702	1.672×10^{-3}	2.128×10^{-3}	0.362
$\pi^{mte3}_{\mathrm{ITP}_1}$	150M	18190.419	1.890×10^{-3}	1.440×10^{-3}	0.367

Table 7 Numerical results of robust trajectories of AP_1

Policy	Training steps	Results		
		m_{2,N_2} , kg	$\Delta r_{2,N_2}$, km	$\Delta v_{2,N_2}$, m/s
$\pi^{unp}_{\mathrm{AP}_1}$	5M	18096.409	0.088	17.374
$\pi^{st}_{\mathrm{AP}_1}$	5M	18039.286	0.146	13.493
$\pi^{obs}_{\mathrm{AP}_1}$	5M	18038.346	0.401	7.808
$\pi^{ctr}_{\mathrm{AP}_1}$	5M	18033.873	0.067	6.007
$\pi^{mte1}_{\mathrm{AP}_1}$	5M	18015.999	0.197	0.020
$\pi^{mte3}_{\mathrm{AP}_1}$	5M	18004.570	0.275	19.697

The velocity increment magnitudes over the robust trajectories are plotted in Fig. 6, where only the increments in the scenario *ctr* are given. In Fig. 6, it can be seen that the applied velocity increments are noticeably lower than the respective maximum admissible magnitudes, which is a distinctive feature of the robust trajectory compared to the optimal one in scenario unp. This is because the trained robust policies tend to leave margins to compensate for uncertainties that may arise during the mission. The control pattern also leads to more propellant consumption along robust trajectories compared to the optimal trajectory in *unp*.

Table 8 Numerical results of robust trajectories of ITP_2

Policy	Training steps	Results			
		m_{3,N_3} , kg	$\Delta r_{3,N_3}$, AU	$\Delta v_{3,N_3}$, VU	$\Delta v^{orb}_{3,N_3}$, km/s
$\pi^{unp}_{\mathrm{ITP}_2}$	300M	15093.566	4.529×10^{-4}	2.512×10^{-3}	0.366
$\pi^{st}_{\mathrm{ITP}_2}$	250M	14971.738	4.243×10^{-4}	1.562×10^{-3}	0.252
$\pi^{obs}_{\mathrm{ITP}_2}$	250M	14995.505	3.369×10^{-4}	1.403×10^{-3}	0.326
$\pi^{ctr}_{\mathrm{ITP}_2}$	250M	14978.850	1.568×10^{-4}	1.513×10^{-3}	0.242
$\pi^{mte1}_{\mathrm{ITP}_2}$	300M	14914.782	5.013×10^{-4}	3.778×10^{-3}	0.198
$\pi^{mte3}_{\mathrm{ITP}_2}$	300M	14894.131	-1.115×10^{-4}	3.618×10^{-3}	0.191

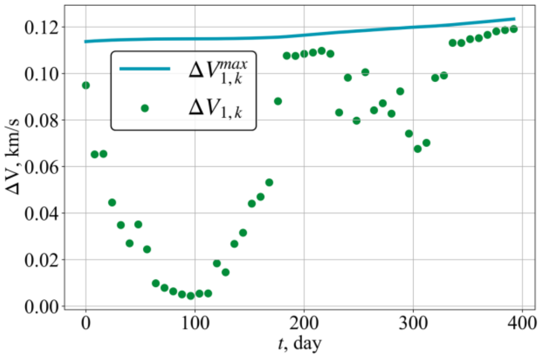
Table 9 Numerical results of robust trajectories of AP_2

Policy	Training steps	Results		
		$m_{4,f}$, kg	$\Delta r_{4,f}$, km	$\Delta v_{4,f}$, m/s
$\pi^{unp}_{\mathrm{AP}_2}$	5M	14957.666	0.017	0.0
$\pi^{st}_{\mathrm{AP}_2}$	5M	14873.106	0.117	0.0
$\pi^{obs}_{\mathrm{AP}_2}$	5M	14880.164	0.093	0.0
$\pi^{ctr}_{\mathrm{AP}_2}$	5M	14878.219	0.165	0.0
$\pi^{mte1}_{\mathrm{AP}_2}$	5M	14823.910	2.496	10.946
$\pi^{mte3}_{\mathrm{AP}_2}$	5M	14813.001	4.270	27.455

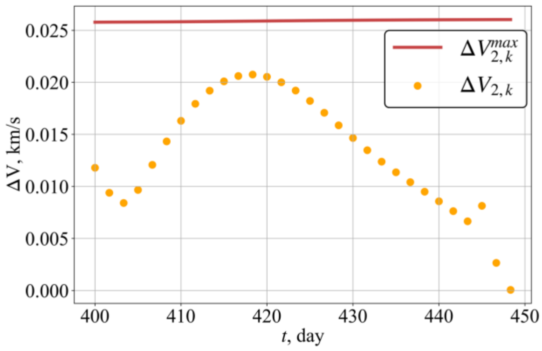
C. Monte-Carlo simulation

Next, 1000 MC simulations are performed in each scenario to validate the closed-loop performance of the trained robust polices. The MC is conducted in each phase, where the end states of the preceding phase are the initial condition for the following one. Meanwhile, we assume that the desired GA maneuver can be achieved if the end position error in AP_1 is below 20 km, otherwise the mission will fail. This error bound is just intended to quantify the performance of a disturbed spacecraft in the APs under the guidance of the robust polices. The obtained numerical results are summarized in Table 10 - Table 13, where success rate (SR) is the percentage of solutions that satisfy the constraints of each phase. For the ITPs, the constraints are $\Delta v^{orb}_{1,N_1} < \Delta v^{orb}_{1,\max}$ in ITP_1 and

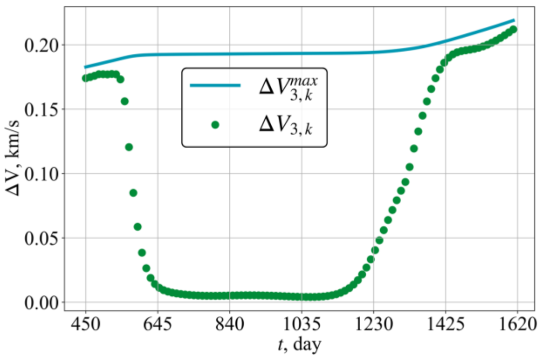
$\Delta v_{3,N_3}^{orb} < \Delta v_{3,\max}^{orb}$ in ITP$_2$, while for the APs, they are $\Delta r_{2,N_2} < 20$ km in AP$_1$ and $\Delta r_{4,f} < 20$ km in AP$_2$.



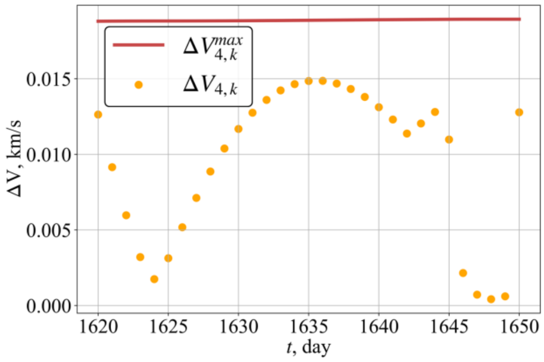
(a) The velocity increment for the ITP$_1$



(b) The velocity increment for the AP$_1$



(c) The velocity increment for the ITP$_2$



(d) The velocity increment for the AP$_2$

Fig. 6 The velocity increment $\Delta V_{i,k}$ over the mission time in scenario *ctr*

Table 10 MC results of the robust policies for ITP$_1$

Policy	m_{1,N_1}, kg		$\Delta r_{1,N_1}$ / AU, 10^{-3}		$\Delta v_{1,N_1}$ / VU, 10^{-3}		$\Delta v_{1,N_1}^{orb}$, km/s		SR, %
	mean	std	mean	std	mean	std	mean	std	
$\pi_{\mathrm{ITP}_1}^{st}$	1824.687	4.514	1.889	0.909	3.345	1.244	0.418	0.025	96.5
$\pi_{\mathrm{ITP}_1}^{obs}$	18265.649	0.026	2.525	0.008	2.378	0.014	0.463	0.001	99.9
$\pi_{\mathrm{ITP}_1}^{ctr}$	18210.055	5.898	1.289	0.453	1.144	0.638	0.402	0.024	98.9
$\pi_{\mathrm{ITP}_1}^{mte1}$	18201.913	24.182	1.479	0.518	1.978	0.578	0.404	0.066	82.8
$\pi_{\mathrm{ITP}_1}^{mte3}$	18195.826	28.243	1.688	0.680	1.717	1.116	0.412	0.085	80.2

Table 11 MC results of the robust policies for AP$_1$

Policy	m_{2,N_2}, kg		$\Delta r_{2,N_2}$, km		$\Delta v_{2,N_2}$, m/s		SR, %
	mean	std	mean	std	mean	std	
$\pi_{\mathrm{AP}_1}^{st}$	18024.019	25.996	2.210	4.831	15.473	9.477	97.9
$\pi_{\mathrm{AP}_1}^{obs}$	18038.311	0.402	1.008	0.449	7.809	0.147	99.9
$\pi_{\mathrm{AP}_1}^{ctr}$	18025.594	13.400	0.429	0.333	6.087	1.399	99.9
$\pi_{\mathrm{AP}_1}^{mte1}$	18011.682	22.330	1.616	6.389	20.631	10.284	97.5
$\pi_{\mathrm{AP}_1}^{mte3}$	17994.037	35.447	4.818	11.371	18.577	5.673	87.4

From these statistical data, it can be seen that the trained robust policies can deal with the considered uncertainties quite well. In scenarios *st*, *obs*, and *ctr*, the EP spacecraft with the trained robust policies can satisfy the position error constraints for the GA maneuver and terminal rendezvous with the SR exceeding 90% or even 95%. For the extreme scenarios *mte*1 and *mte*3, the spacecraft can still achieve the mission objectives with the SRs of more than 80% and 70%, respectively. For the final mass, the spacecraft consumes the most propellant in *mte*3, which is about 3.32% more propellant than in scenario *unp*.

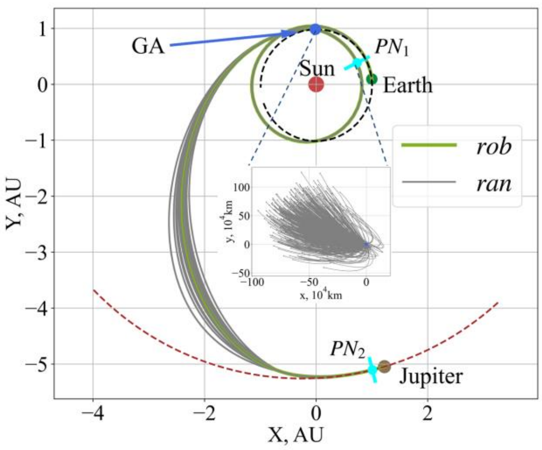
Table 12 MC results of the robust policies for ITP_2

Policy	m_{3,N_3}, kg		$\Delta r_{3,N_3}$ / AU, 10^{-3}		$\Delta v_{3,N_3}$ / VU, 10^{-3}		$\Delta v^{orb}_{3,N_3}$, km/s		SR, %
	mean	std	mean	std	mean	std	mean	std	
$\pi^{st}_{\mathrm{ITP}_2}$	14959.999	17.943	0.726	0.300	1.678	0.547	0.262	0.017	97.9
$\pi^{obs}_{\mathrm{ITP}_2}$	14994.728	0.751	0.497	0.209	1.395	3.664	0.328	0.006	99.4
$\pi^{ctr}_{\mathrm{ITP}_2}$	14970.967	11.850	0.674	0.356	1.469	0.357	0.256	0.201	99.9
$\pi^{mte1}_{\mathrm{ITP}_2}$	14912.129	22.105	1.708	2.216	3.889	1.290	0.284	0.241	85.7
$\pi^{mte3}_{\mathrm{ITP}_2}$	14881.030	76.259	1.607	2.759	3.549	1.484	0.300	0.305	77.8

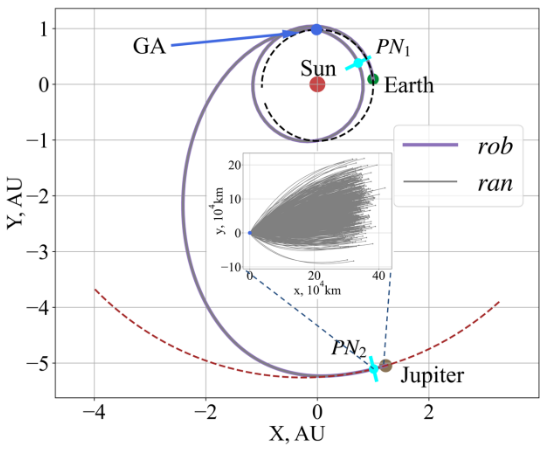
Table 13 MC results of the robust policies for AP_2

Policy	$m_{4,f}$, kg		$\Delta r_{4,f}$, km		$\Delta v_{4,f}$, m/s		SR, %
	mean	std	mean	std	mean	std	
$\pi^{st}_{\mathrm{AP}_2}$	14850.917	22.666	1.509	4.060	1.281	5.129	97.5
$\pi^{obs}_{\mathrm{AP}_2}$	14871.889	6.445	0.356	3.932	0.032	0.512	99.5
$\pi^{ctr}_{\mathrm{AP}_2}$	14852.272	28.445	2.358	6.230	0.161	2.237	95.7
$\pi^{mte1}_{\mathrm{AP}_2}$	14817.455	32.767	8.446	13.956	15.776	30.352	80.7
$\pi^{mte3}_{\mathrm{AP}_2}$	14796.096	42.021	10.185	14.310	30.704	9.853	72.2

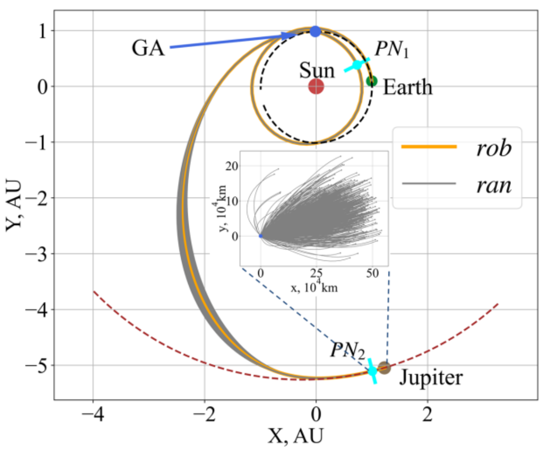
Besides, Table 14 gives three training cases for *mte*3 in ITP_1 with different discount factors κ and coefficients c_3 of reachability constraints for comparison. For Case 1, the factor $\kappa_1 = 1$ and $c_3 = 0$, indicating that the initial nominal state of AP_1 is not modified and reachability constraint is not considered in training. As for the other, $\kappa_1 = 1$ and c_3 is the same as that in Table 5 for Case 2, and $\kappa_1 = 0.6$, $c_3 = 0$ for Case 3. During training, classical ZEM/ZEV guidance law [34] ($K_R = 6$, $K_R = -2$) is adopted to accurately assess the reachability from the end states of RL-based ITP trajectories to the target state of the AP in an unperturbed AP. This can help us to make a better initial assessment of the trained robust policies of the ITPs without the need for time-consuming training of the APs. For the end states of ITP_1 in Table 14 and those from policy $\pi^{mte3}_{\mathrm{ITP}_1}$, the SRs of satisfying the constraint $\Delta r_{2,N_2} < 20$ km in the unperturbed AP$_1$ are 0, 83.6%, 35.4%, and 97.7%, respectively. Thus, it is seen that the introduction of the discount factor and reachability constraints, especially the latter, significantly improves the reachability from the end states of the ITP under uncertainties to the target state of the AP.



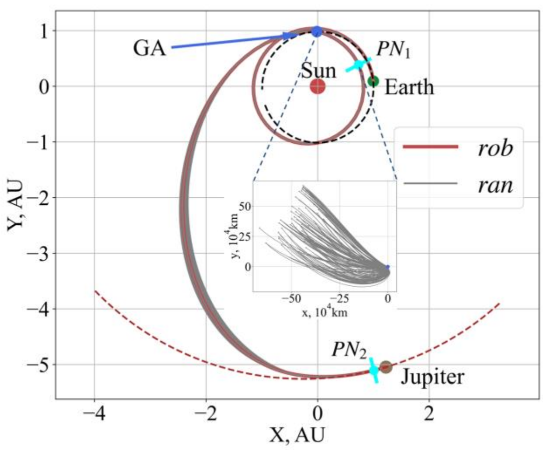
(a) Random trajectories in scenario *st*



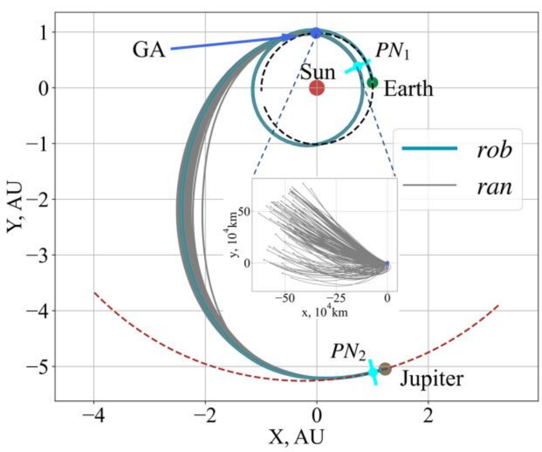
(b) Random trajectories in scenario *obs*



(c) Random trajectories in scenario *ctr*



(d) Random trajectories in scenario *mte*1



(e) Random trajectories in scenario *mte*3

Fig. 7 Monte Carlo simulations of the trained robust policies

MC simulation trajectories in each scenario are shown in Fig. 7. In Fig. 7, it can be noticed that there exist wider dispersions in the middle part of the random trajectories of the ITPs, which tend to decrease or even diminish toward the end to meet the terminal constraints. The position differences between the spacecraft and the target object over the APs are also plotted for better illustration in Fig. 7. From the locally magnified view of the position differences, it is clear that the spacecraft moves toward the target during the APs.

Table 14 Numerical results of the policies by different discount factors and reachability index coefficients

Parameter	Training steps	m_{1,N_1}, kg		$\Delta r_{1,N_1}$, AU		$\Delta v_{1,N_1}$, VU		$\Delta v_{1,N_1}^{orb}$, km/s	
		mean	std	mean	std	mean	std	mean	std
Case1	200M	18333.853	52.121	0.012	0.013	0.014	0.011	1.384	0.873
Case2	200M	18323.054	27.446	0.002	0.002	0.005	0.002	0.519	0.119
Case3	200M	18172.850	23.138	0.005	0.006	0.006	0.003	0.831	0.470

Finally, to further verify the generalization ability of the trained robust policies, they are applied to extended cases, where the uncertainty parameters are 1.5 times those adopted in training. Only the uncertain scenarios *st*, *obs*, and *ctr* are considered, as the original scenarios of *mte*1 and *mte*3 have already been assumed to include at least one missed thrust event in each phase. Similarly, 1000 MC simulations are performed, and the relevant numerical results are presented in Table 15 - Table 18. It can be seen that the performance of the trained robust policies shows slight degradation compared to the original cases, but it is still promising. The position error constraints for the GA and terminal rendezvous can be satisfied with SRs exceeding 90% and 84%, respectively. As for the extended cases with larger uncertainty parameters, the performance of trained robust policies will deteriorate further, as an inherent limitation of ML methods.

Table 15 MC results of the robust policies in the extended cases for ITP_1

Policy	m_{1,N_1}, kg		$\Delta r_{1,N_1}$ / AU, 10^{-3}		$\Delta v_{1,N_1}$ / VU, 10^{-3}		$\Delta v_{1,N_1}^{orb}$, km/s		SR, %
	mean	std	mean	std	mean	std	mean	std	
$\pi_{\mathrm{ITP}_1}^{st}$	18225.309	4.218	2.377	1.232	3.742	1.717	0.440	0.044	84.0
$\pi_{\mathrm{ITP}_1}^{obs}$	18265.650	0.038	2.525	0.011	2.378	0.020	0.464	0.001	99.9
$\pi_{\mathrm{ITP}_1}^{ctr}$	18210.941	8.343	1.511	0.659	1.436	0.839	0.415	0.036	93.0

With the above data, it can be concluded that the trained robust policies can deal with the uncertainties well, especially in satisfying the desired accuracy for GA maneuver and terminal rendezvous. The trade-off factor technique and the reward design with reachability constraint significantly improve the reachability from the end states of RL-based ITP trajectories in the considered uncertain scenarios to the target state of the APs.

Table 16 MC results of the robust policies in the extended cases for AP_1

Policy	m_{2,N_2}, kg		$\Delta r_{2,N_2}$, km		$\Delta v_{2,N_2}$, m/s		SR, %
	mean	std	mean	std	mean	std	
$\pi^{st}_{AP_1}$	18002.859	42.674	6.200	10.932	21.557	20.609	90.1
$\pi^{obs}_{AP_1}$	18038.347	0.558	0.398	0.182	7.808	0.050	99.9
$\pi^{ctr}_{AP_1}$	18014.931	27.151	1.093	3.884	6.448	3.147	98.7

Table 17 MC results of the robust policies in the extended cases for ITP_2

Policy	m_{3,N_3}, kg		$\Delta r_{3,N_3}$ / AU, 10^{-3}		$\Delta v_{3,N_3}$ / VU, 10^{-3}		$\Delta v^{orb}_{3,N_3}$, km/s		SR, %
	mean	std	mean	std	mean	std	mean	std	
$\pi^{st}_{ITP_2}$	14945.392	31.732	1.022	0.473	1.912	0.996	0.274	0.029	89.8
$\pi^{obs}_{ITP_2}$	14993.870	1.040	0.642	0.297	1.389	0.046	0.332	0.011	99.4
$\pi^{ctr}_{ITP_2}$	14963.590	20.088	1.043	0.568	1.346	0.460	0.276	0.036	96.5

Table 18 MC results of the robust policies in the extended cases for AP_2

Policy	$m_{4,f}$, kg		$\Delta r_{4,f}$, km		$\Delta v_{4,f}$, m/s		SR, %
	mean	std	mean	std	mean	std	
$\pi^{st}_{AP_2}$	14825.589	37.964	4.718	10.546	6.220	15.245	84.3
$\pi^{obs}_{AP_2}$	14864.755	12.262	1.357	6.929	0.975	5.262	97.1
$\pi^{ctr}_{AP_2}$	14835.223	36.860	5.626	11.356	6.093	17.038	87.6

V. Conclusion

This paper presents an RL-based multi-phase method for robust low-thrust GA trajectory design. To alleviate the complexity due to the interior-point constraints induced by GA, the low-thrust trajectory legs before and after GA are segmented into ITPs and APs, respectively. In each phase, the MDPs are modeled separately for the various considered uncertainties. For the problem of failing to reach the target of the AP due to the end state deviations of the ITP, the proposed trade-off factor technique and reward design with reachability constraint significantly improve the reachability from the end states of the RL-based ITP trajectories to the target of the AP. As well, the R-ZEM/ZEV method is applied to generate robust guidance laws to compensate for uncertainties over the AP to achieve the high approaching accuracy. The promising results validate that the

RL-based multi-phase robust GA trajectory design method performs well in low-thrust exploration missions with intermediate GA, and it also provide an alternative to the trajectory design problems with interior-point constraints. Besides, since this study focuses more on robust trajectory design problem, autonomous navigation methods were not considered, especially for the APs, it should be investigated in the future.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 12372046), the Natural Science Foundation of Jiangsu Province (No. BK20220130) and the Fundamental Research Funds for the Central Universities (No. NE2025008). A portion of this paper was presented as Paper IAC-23, C1, IP, 8, x76781 at the 74th International Astronautical Congress, Baku, Azerbaijan, 2-6 October 2023.

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